Prague University of Economics and Business

Faculty of Informatics and Statistics

SUPPORT IN REPORTING AND REGISTRATION OF BOSNIAN EMIGRANTS FOR THE EMBASSY OF BOSNIA AND HERZEGOVINA IN PRAGUE

BACHELOR THESIS

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Prohlášení

Prohlašuji, že jsem bakalářskou práci Support in reporting and registration of Bosnian emigrants for the Embassy of Bosnia and Herzegovina in Prague vypracovala samostatně za použití v práci uvedených pramenů a literatury.

V Praze dne 9. května, 2021 ....

Ivona Josipović

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Abstrakt

Hlavním cílem této bakalářské práce, která se zaměřuje na analýzu potřeb a problémů Velvyslanectví Bosny a Hercegoviny v Praze, je vytvoření aplikace pro přidávání a editaci záznamů a budování dynamického dashboardu podle stanovených potřeb. Hlavním cílem je vytvořit SW aplikaci pro registraci a hlášení emigrantů. V teoretické části autor vysvětluje základní pojmy vizualizace dat, dashboardy a principy jejich vytváření. Tato část obsahuje také úvodní kapitolu o Velvyslanectví Bosny a Hercegoviny v Praze. Praktická část definuje potřeby a kritéria pro vytvoření systému a dashboardu, které jsou stanovené na základě rozhovoru s pracovníky Velvyslanectví Bosny a Hercegoviny v Praze. Hlavním přínosem a výsledkem této části bude funkční aplikace a dashboard vytvořený s doporučenými principy z teoretické části práce. V poslední části testování uživatelů ověří funkčnost aplikace a dashboardu a to, zda je uživatelé mohou snadno využít.

Klíčová slova

dashboard, data, emigranti, velvyslanectví.

Abstract

The main goal of this bachelor's thesis, which focuses on analyzing the needs and problems of the Embassy of Bosnia and Herzegovina in Prague, is to create an application for adding and editing records and building a dynamic dashboard according to the established needs. The main goal is to create a SW application for emigrant registration and reporting. In the theoretical part, the author explains the basic concepts of data visualization, dashboards, and its creation principles. This section also includes an introductory chapter on the Embassy of Bosnia and Herzegovina in Prague. The practical part defines the needs and criteria for creating a system and dashboard, determined by an interview with the staff of the Embassy of Bosnia and Herzegovina in Prague. The main benefit and result of this part will be a functional application and dashboard, created with the recommended principles from the thesis's theoretical part. In the last part, user testing verifies the application and dashboard functionality and whether users can easily utilize them.

Keywords

dashboard, data, embassy, emigrants.

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Introduction

Over the course of the last few years, the amount of generated data that should be analyzed and represented has been rapidly increasing. Accordingly, possibilities for data processing and visualization are growing. Indeed, new trends emerge every year to gain insight on important information both quickly and effectively. The type of data visualization that is arising the most is a dashboard. The use of dashboards by companies is increasing as they have become an essential component in every organization. Due to its increasing popularity, dashboards are constantly optimized and improved upon to avoid severe damage it could otherwise cause. Efficiently displaying visuals on a dashboard is a serious and widespread sphere. As such, there is a significant number of authors dealing with this topic.

Reason for choosing the topic: Being a daughter of a Bosnian diplomat that worked for the Embassy of BiH in Prague, the author had the opportunity to observe its organization and lack of digitalization. The output of this thesis can be of help for their development from papers to the digital world. The secondary reason for choosing this topic is being interested in data visualization and its parts.

Objectives: The aim of this thesis is to create a software solution for the Embassy of Bosnia and Herzegovina in the Czech Republic, which would add new records and an output dashboard with analyzed data. The minor goal is to help the staff of BiH embassies with the registration of emigrants of Bosnia and Herzegovina and their analysis, as there is no system for this need. Another secondary goal is to clarify principles for creating and designing dashboards and applying these directions to create a new dashboard for the Bosnian Embassy's needs.

Methods for achieving the goal: In the first part of the thesis, the author analyzed relevant sources and pulled out relevant content, which was then used to create the thesis's theoretical part. In the practical part, the author first analyzed the embassy's current situation and defined their needs and criteria. To create the outputs for the practical part, author used the practices and theories defined in the above theoretical part. To test the functionality of the app and dashboard, the embassy's staff was used. They were provided with a questionnaire to help with evaluation and reviews.

1 Introduction to Data Visualization

The chapter of Introduction to Data Visualization, focuses on the defining data visualization and its stages of creation. The following subchapters will be about its types and usage across industries.

1.1 What is data visualization

"It's the study of how to represent data by using a visual or artistic approach rather than the traditional reporting method" (YUK et al., 2014).

As the name says, data visualization gives a visual insight into our data. It helps in working with a large amount of data, e.g., thousands of rows or with complicated relations, providing relevant information. The word that describes it the best would be simplifying. As a result of this process, there is no need to search for the answer in random tables of data. Instead, it is enough to look at the visuals, to find the needed information. It is more efficient, time-saving, and overall gives better results than old-time reporting. It is essential to understand its importance. Companies, especially ones operating with websites and apps, have the most significant need of data visualization. Such programs generate a large amount of data that contains valuable pieces of information within.

Since the rise of social media and apps, data has been accumulating in enormous sizes. They are generated every moment and contain many vital pieces of information that are hard to see all at once. Hence, many companies invest in data visualization tools and systems because they allow them to read and use a significant amount of data. One can explain this problem with a metaphor of a blindfolded man attempting to go to work. This person will not see the crucial things and will end up having an accident. This metaphor can be applied to business as well. Lack of good and vital information can be a crashing experience in the future. Companies invest in clear visualization tools to gain insights into current performance or future problems and opportunities. (YUK et al., 2014)

When talking about the visualization itself, one question arises: does it look appealing? When creating the visualizations, one should focus on making them visually pleasing to those who will read them. To do so, the creator must choose the right colors, fonts, and visuals themselves. Not every graph or chart can be used for all types of data. More about this topic will be discussed in Chapter 3. However, the most critical is to focus and select correct items and parts to gain essential information. Mico Yuk and Stephanie Diamond (YUK et al., 2014) talk about the traits of data viz, and use the following:

- ***"Useful:*** *People use it on a regular basis and can make relevant decisions by viewing all the information that they need in one place.*
- ***Desirable:*** *It is not only easy to use but also pleasurable to use.*
- ***Usable:*** *People who use it can accomplish their goals quickly and easily.*"

1.2 Steps of visualization process

When talking about data viz as a process, we must also recognize that it is a long one. It starts with raw data that is later turned into visuals, like a dashboard. Ben Fry (Fry, 2007) explains the steps of visualization processes in his book Visualizing data (Figure 1.1).

The whole process begins with obtaining/collecting the data. It can be from various sources or just one, such as a disc or network and can appear in any format. The main question for this part would be: what kind of data do we want and how much of it do we need? Nevertheless, the main question throughout the whole process is: what do we want to accomplish, what needs to be answered, and how do we make it fast?

After collecting data, we need to parse it, meaning that we need to edit it. Editing data into categories and giving it structure gives us a good platform for the next steps and makes it easier to work with.

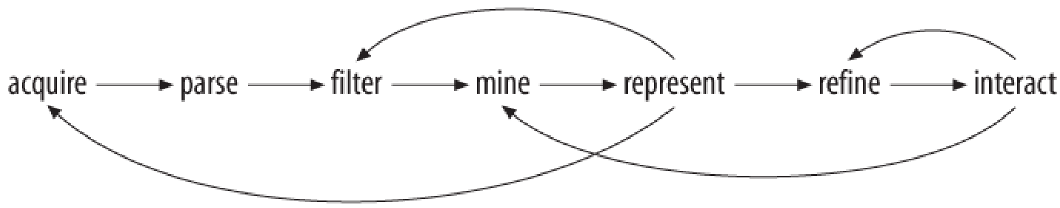


Figure 1.1 Stages of visualization process, Source: (Fry, 2007)

The next step should be oriented to the above questions because we need to filter data. It is one of the most important parts of the process. One should not work with enormous amounts of data, but rather the ones that can give something. Not only do we then have what we need and want, but it reduces the data to only that which is useful. Once the data is filtered, we can continue to mining. This step mostly involves performing mathematical and statistical operations on data. It can vary from simple operations, like finding the average or max value, but can also turn into more complicated formulas.

After the mining step, data is usually ready to be represented with some basic visuals. This step is most crucial because the dataset gets its form in which it will be used and shown. They can be in the states of trees, lists, and so on.

Once the basics are done, one can continue on to the refining stage. At this stage, in which basics turn into pleasing visuals. Attention is given to graphically editing data, using color theory and chart theory, to make the dashboard design appealing to the reader. The goal is to end up with a good-looking dashboard.

The process ends with the interaction stage. In this step, users gain control and can use and analyze data for their own needs. It gives a user the possibility to change the viewpoint, which can invoke some changes in the previous steps, such as adjusting data to function in different perspectives.

1.3 Data visualization content types

Data visualization is divided into more types, but the two most popular are dashboards and infographics. Using tables, charts, and images, they provide data presentation for better understanding and gaining information. When we talk about these two types, it is essential to mention that they are very different, even though they have the similar goals and principles. (Iliinsky et al., 2011)
In the book, Data Visualization for Dummies, Mico Yuk and Stephanie Diamond (Yuk et al., 2014) introduced some of the criteria to distinguish these two types:

- **Quantity of data:** mostly data visualizations use large quantities of data, with frequent changes and real-time data updates. This makes them more flexible. In contrast, infographics contain data that does not get updated or changed as often and is more static.

- **Aesthetics:** This criterion is more about art and graphic design. If information is represented with artistic and well-done design, it is probably meant to be an infographic.

- **Interactivness:** If a graphic is meant to tell a story through visualization, it uses static data that is not meant to be changed often, and in this case, we talk about infographics. If the story changes every time we add or update data, we can say that those are interactive graphics, meaning data visualization.

- **Illustrations:** Infographics tend to use illustrations and art graphics to represent data, while data visualization is more number oriented.

Example of infographics (Figure 1.2**) vs data visualization dashboard (**Figure 1.3**):**

Figure 1.2 Infograph example, Source: (Mahnoor, 2020)

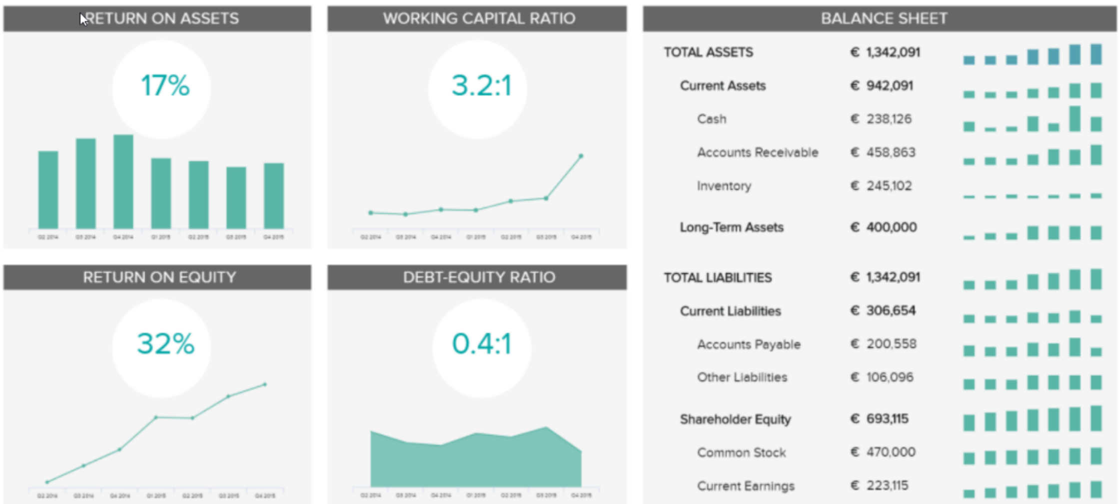


Figure 1.3 Financial Performance Dashboard, Source: (Durcevic, 2020)

When creating data visualization to showcase events and stories, we have the option to use as many content types as desired. It is only essential to choose the right visuals for the suitable types of data.

The following table contains some of the content types (Table 1.1), additional information about content types will be explained in Chapter Three.

Table 1.1 Content Types, Source: YUK et al., 2014

Content types	
Chart	Uses two axes (x and y) to create visuals for data
Diagram	It explains how things work
Timeline	Chronologically shows changes using graph
Template	Guidline for the user when creating or filling in something
Checklist	Keeping track of tasks on a list which can be crossed off when finished
Flow chart	Set of steps to help a user understand how something works
Metaphor	Shows differentiation between two things using comparison methods

1.4 Where to use it

Visualization is all around us. It can be a menu in a restaurant or a free flyer from the supermarket. Usage is widely ranged across many sectors, such as marketing and sales, politics, human resources, healthcare.

Data visualization is almost mandatory in the sector of marketing and sales. Since we live in a digital world, most of the revenue comes from the web and apps. Consequently, marketers and salespeople need to pay attention to important details to make good plans for future activities. Using charts, graphs, and trend lines over time can improve overall predicting.

Another sector of great usage is in finance, where it helps with salary management, tracking investments' performance and price movements in general. The most used chart in this sphere would be candlestick. It monitors price movements and shows vital information like stocks, bonds, securities, and others. Politics and healthcare have a similar need for data visualization. This sector mainly uses geographical chart map to represent, e.g., voting during elections or mortality rate in healthcare by regions. Data visualization is further used in the science field to properly analyze and represent experimental data. (Brush et al., n.d.)

2 Introduction to Dashboards

2.1 What are dashboards

"*A dashboard is a visual interface that provides at-a-glance views into key measures relevant to a particular objective or business process."* (Alexander, 2008)

Alexander Michael (Alexander, 2008) defines dashboard's three main attributes:

- **graphical:** they are usually in a visual form made out of graphics, allowing it to monitor key trends
- **relevant:** each dashboard has its own goal, and they typically show only data that is relevant for achieving those goals
- **predifined:** end-user does not need to work on his analysis because dashboards use predefined conclusions and results to the purposes mentioned above

According to Gartner: "*Dashboards are a reporting mechanism that aggregate and display metrics and key performance indicators (KPIs), enabling them to be examined at a glance by all manner of users before further exploration via additional business analytics (BA) tools. Dashboards help improve decision making by revealing and communicating in-context insight into business performance, displaying KPIs or business metrics using intuitive visualization, including charts, dials, gauges and "traffic lights" that indicate the progress of KPIs toward defined targets."* (Gartner, n.d.)

The reason for dashboards being so widely used is that they give an immediate picture of the state of many processes. Everything we need to know is in one place, making it very convenient to use and easy to read. It is a tool that can efficiently support more than one data source and provides real-time monitoring. Real-time monitoring means that it continually updates new data and changes accordingly, which is time-saving, and other business processes can run smoother.

Proof that usage of dashboards is growing is shown in the following graph by Select Hub (Figure 2.1). *"The results demonstrate a higher need for basic features, with advanced analytics considered as extras."* (Adair, n.d.). Almost 90% of respondents picked out dashboards, showing that it is one of BI's most essential features. The second vital feature is visualizations, according to more than 80% of respondents.

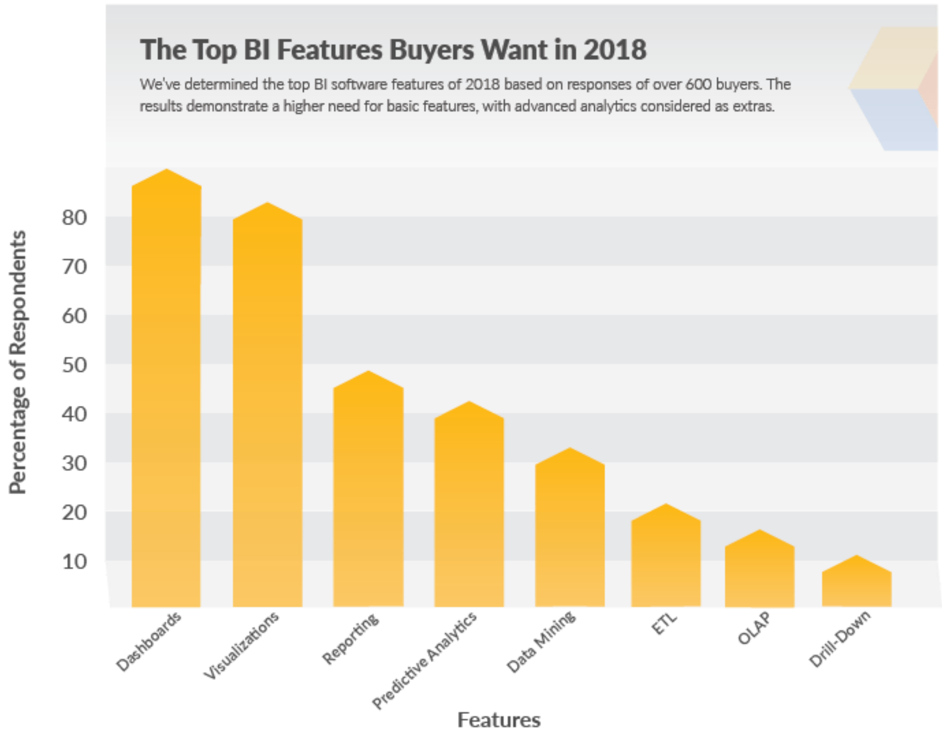


Figure 2.1 The Top BI Features Buyers want in 2018, Source: (Adair, n.d.)

Implementing a dashboard system into business brings many benefits to the company. Since data is real-time, it gives insight into the current situation and performance, allowing a business to make decisions quickly and safely. Another relevant benefit is its user-friendly interface. They usually fit on one page, where all important things are in one place. This system often plays a role in motivating employees, who do not have to lose time on their analysis, but can read the dashboard pieces correctly. (Ramos, 2016a)

In cases where a business wants to invest in only one type of dashboard, it should put on paper all the needs, style of data, and preferences. However, one should not worry about missing some essential features, because as said above: all dashboards have standard settings, such as: customization, an offer of graphs and charts, user-friendly interface, color theory and fonts, add-ins for databases, graphics, platform support and options for cloud-based or open-source service. (Ramos, 2016a). More about some of the features will be explained in Chapter Three.

2.2 Key Performance Indicators

Dashboards are typically built on measures or key performance indicators. "*Key performance indicators (KPIs) are those indicators that focus on the aspects of organizational performance that are the most critical for the current and future success of the organization.*" (Parmenter, 2015)

They represent the performance of crucial every-day processes and operations within the business. They show both problematic or success parts that need to be seen and acted upon. (Alexander, 2008).

The following table, taken from David Parmenter's book, Key Performance Indicators: Developing, Implementing and Using Winning KPIs, defines the top seven characteristics of KPIs. (Table 2.1)

Table 2.1 Characteristics of KPIs, Source: Parmenter, 2015

Characteristics of KPIs (edited)	
Nonfinancial	Those types of measures are not expressed in e.g., Euros, CZK
Timely	Measured in real-time (daily, weekly)
Used by	Top management and CEO, or also individuals
Simple	Everyone understands it and what indicators it shows
Team-based	Can be used by one or more teams
Impact	It is an excellent benefit to the whole business
Downsides	Performance measure should be tested before turning into KPI, but if not done correctly, it can lead to wrong answers

When defining KPIs for business, companies should take their time and follow some of the tips (Walczak, 2014):

- Focusing on the right indicators, that would benefit in achieving goals
- Focus on understanding metrics completely, for the better end results
- Simplicity is the key
- Relation with company's goals

2.3 Types of dashboards

This subchapter aims to introduce types of dashboards to the reader. An example and specifications will be provided for each class discussed.

Since dashboards offer so much in the way of functions, possibilities for visualization, and data interpretation in today's world, it is tricky to realize that not every type of dashboard can fit everywhere. This can be very hard for companies that consider implementing such a system into their environment. Although each of them is usually intended and specified for one sphere, they do have functions in common. Nevertheless, companies typically decide to incorporate all three types: operational, tactical, and strategic. (Durcevic, 2020)

In addition to these types, which will be analysed more later in the chapter, we can further divide them into additional groupings: free software, on-premise, website, and open-source: (Ramos, 2016b)

Free software is suitable for beginners who want to either start using them for the work or for people who want to learn to build dashboards.

Website dashboards, as the name implies, are used for checking out website performance or online activities. It can be connected with Google Analytics, Pingdom, and other software.

Both of these groups can belong to the open-source type. There is a sheer variety of them, and they can be web or server-based.

Here we can see how Eckerson (2006) classified dashboards into three groups in their presentation. Some of the info from the table (Table 2.2) will be used in the following explanation of each type.

Table 2.2 Types of dashboards, Source: Eckerson 2006

	Operational	**Tactical**	**Strategic**
Focus	Monitoring operations	Optimize process	Execute strategy
Emphasis	Monitoring	Analysis	Collaboration
User	Supervisors	Managers	Executives
Scope	Operational	Departmental	Enterprise
Information	Detailed	Detailed/Summary	Summary
Updates	Daily	Daily/monthly	Monthly/Quarterly
Form	Dashboard	BI Portal	Scorecard

2.3.1 Operational

Operational dashboards are probably the ones that are used the most. They monitor and manage operational processes and activities going on in the business daily using real-time data. This dashboard type is primarily used by different management levels, either junior levels for basic operational processes or by higher supervisors that take care of more significant activities.
Since it contains real-time data, it can quickly point to the possible problem on time, keeping things under control. Thus, they are usually more detailed than other types, and display critical and problematic components for a business in greater detail. Apart from being able to detect a problem ahead, it can also detect opportunities (Table 2.3).

"*Operational dashboards help departments stay proactive and ahead of problems. For example, a manufacturing firm may use an operational dashboard to track products manufactured along with the number of defects, complaints, or returns. This helps in the manufacturing analytics processes – with a dashboard, any problematic changes would be highlighted in real-time.*" (Durcevic, 2020).

Table 2.3 Operational dashboard, Source: Eckerson, 2011; Rasmussen et al., 2009

Characteristics: supports monitoring of business processes and activities. It gives insight into various changes, problems, or opportunities. It is mainly used in the environment, for which it is vital to have constant supervision of processes going on and their changes. It can be used in many spheres, but it fits best in marketing, sales, human resources.
Pros: detects risks or opportunities on time, motivates and improves employees, shows the current situation in the company.
Cons: can mislead manager if there is a small mistake, cannot be trusted 100% for decision making, data does not go through high data validation, there can be a risk with data quality.
Example: Figure 2.2

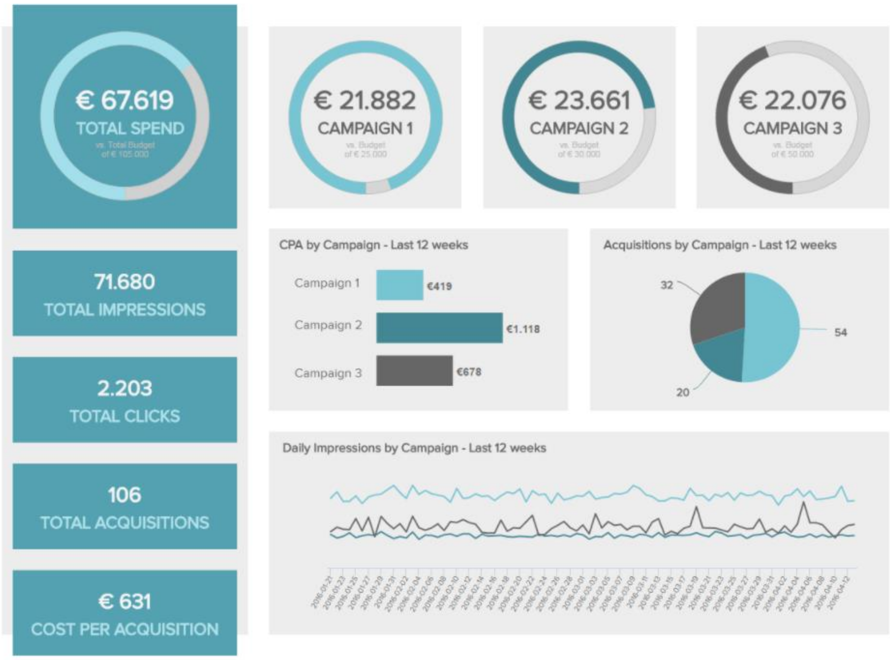


Figure 2.2 Example: Marketing operational dashboard, Source: (Durcevic, 2020)

2.3.2 Tactical

This is the type that is very vital for many future processes and strategies of a company. It uses historical data that is usually updated weekly, and in some cases, daily too. With the help of analyses, it can create trends, targets, and even predictions that can be incorporated into the company's strategic plan and BI. It helps optimize processes within the company, which helps managers know which path to choose and which decisions to make. Visualization is crucial for this type because oftentimes, it uses very complex data that is then presented in an easy way to read and understand. Other than complex, data used for this type can also be detailed or summarised. As said above, it can be part of the company's Business Intelligence system because it has the same goal and purpose: to support decision-making (Table 2.4).

Table 2.4 Tactical dashboard, Source: Eckerson, 2011; Rasmussen et al., 2009

Characteristics: It is a tool to help with the optimization of processes in different departments. Visualization is crucial, so it uses various tools for reporting and analytics. It can be incorporated into BI since it also uses data warehouses and data markets.
Pros: identifies trends, very detailed, user can see it from different dimensions, can be used for statistics and predicting.
Cons: BI infrastructure must be built very good to support systems correctly; metrics and terms should be standardized if the different departments understand and use them differently.
Example: Figure 2.3

Figure 2.3 Example: Financial Performance Dashboard, Source: (Durcevic, 2020)

2.3.3 Strategical

Similar to the tactical type, strategical dashboards use historical data to monitor the company's strategy, based on previous KPI performances compared with the company's goals. It is more complex than other groups because it focuses on a company as a whole and rather than separate parts; hence executives or senior managers are the ones who work with strategical dashboards. Data can be updated monthly or quarterly, but the outputs can also be in years. There is no need for detailed data; summarized ones are sufficient because it is more often used to show performance over time, rather than making decisions quickly. When adequately developed and implemented, they can provide a high-speed and precise image of issues or opportunities, giving executives time and opportunity to act on it. In the end, the most significant output from it is a scorecard that can be generated fast without time-consuming steps and analysis (Table 2.5).

Table 2.5 Strategical dashboard, Source: Eckerson, 2011; Rasmussen et al., 2009

Characteristics: This dashboard is used for maintaining and achieving strategic goals. It monitors the development and movement towards the goal and shows the current state of the situation. It does not usually use real-time data. Hence it works at high speed.

Pros: gives a bird-eye view, defines, monitors, and controls strategic business goals, an excellent tool for executives, can help with discovering risks and opportunities

Cons: takes time to implement it, already defined goals can differ from already made ones

Example: Figure 2.4

Figure 2.4 Example CFO strategical dashboard, Source: (Durcevic, 2020)

3 Principles of building dashboards

This chapter will be about the principles of building dashboards. It will focus on data models and relations between them. Color theory, graphs, and charts will also be discussed in relation to visualization.

3.1 Dimensions and metrics

To understand this subchapter better, it is essential to define dimensions and metrics.

"A business ***metric*** *is a quantifiable measure that businesses use to track, monitor, and assess the effectiveness of business processes."* (Ramos, 2016b). When defining the business dashboard metrics, one should consider tailoring them by different areas of the company. However, it is also vital that it is fitting to the end-users needs because, e.g., the marketing department does not need tracking of employee fluctuation like HR does. It is good to compare defined metrics with business goals and established company benchmarks to achieve and track goals. Some of the examples are costs, ratios and revenue. (Ramos, 2016b)

Dimension can be understood as a category, e.g., department, and each category has its values, like sales, marketing, or HR. It then navigates processes or communications between those departments. It also divides measures into groups and tells how they will be distributed. It is vital to define all dimensions at the beginning of the dashboard development because including new ones can be a rather complicated process. (Alexander, 2008)

Ambler defines two primary ways to select key metrics: general and tailored. The general method means that a few of the fundamental ones are defined and can be used across various departments, companies, and time-periods since they are compatible and comparable with their benchmarks. In contrast, the tailored method believes that firms or units have their own goals and strategies, on which the metrics should then be based and individualized. (Ambler, 2003)

3.2 Color Theory

It is safe to say that color plays a huge role in data visualization. It is an essential component in everyday life, so it is only natural to assume that it will be necessary for visualization too. It is the first thing that the human eye catches, so it is crucial to have color adequately chosen and incorporated. Otherwise, it can make things look messy and confusing and even hard to read. For example, if someone wants to showcase data in a pie chart and decides to use various shades of one color, chances are that it will be hardly understandable and difficult to distinguish. When selecting color palettes, one should reflect upon their data to make the most impacting visuals.

When talking about color, we need to think about their variables, defined by Albert H. Munsell, creator of Munsell color theory: lightness, hue, and saturation (Figure 3.1). Lightness can be described with a range between white and black, saturation between dull and bright, while hue is made of red, orange, yellow, green, blue, and purple. (Krystian, 2016)

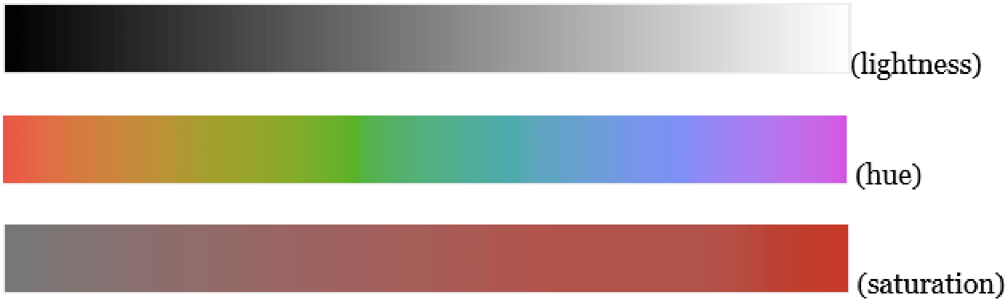


Figure 3.1 Color variables, Source: (Krystian, 2016)

There are three types of color palettes used for representing data, (Yi, 2019):

- *Qualitative palettes*
- *Sequential pallets*
- *Diverging pallets*

3.2.1 Qualitative palettes

Categorical data is represented with qualitative palettes because it can have distinct labels without intrinsic ordering. If it can be ordered in a way, it does not belong to the nominal type but the other groups. Some of the categorical variables are gender, hair color, or binary variable.

When creating and working with this group, one must use distinct colors. There is a golden rule: limit to the size of ten or even fewer colors. Otherwise, it can cause discomposure. If we have more values than colors in a palette, we should do a regrouping of data and merge some of them, creating new categories. This will improve visualization, as well as be easier for the reader to comprehend. Hue is the color variable that helps with distinctiveness, so it is more used for this purpose. Meanwhile, lightness and saturation can be utilized for other variations. (Yi, 2019)

Example: DO and DON’T

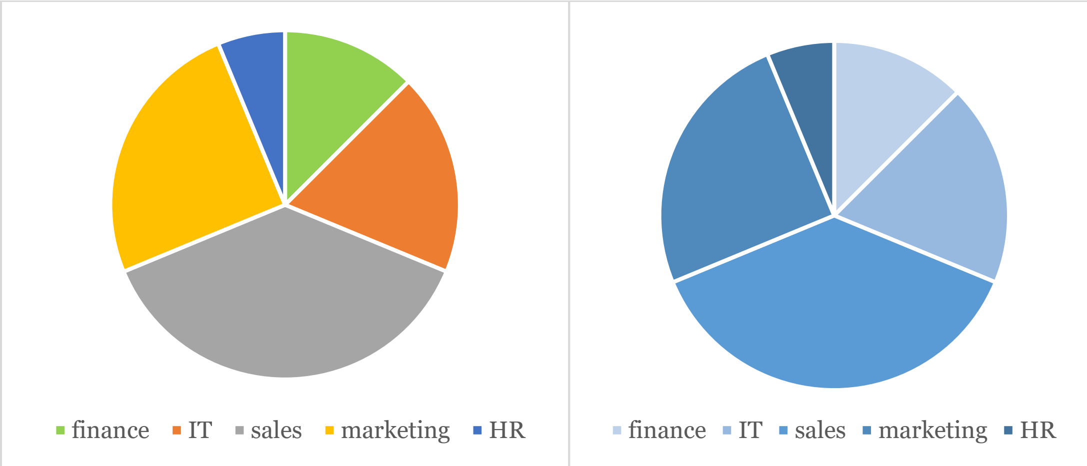


Figure 3.2 Example DO, Source: Author

Figure 3.3 Example DON’T, Source: Author

We can see why it is important to use different colors for representing nominal data. The first pie chart (Figure 3.2) is easy to read and distinguishes categories without showing values. In contrast, the second pie chart (Figure 3.3) ineffectively conveys information, even with a legend present.

3.2.2 Sequential palettes

This type works best for the ordinal values. That means that those values can have clear order, usually represented with numbers, for example, from 1 to 5 or 0 to 100%. Color variables used here are mostly lightness and hue or a combination. We can say that it means that we choose one color and use its various shades. We use lighter shades to reflect lower values, while using darker shades to show higher values. (Figure 3.4). Of course, background plays a role too, and requires a reverse in shades when the background is dark. (Yi, 2019)

Figure 3.4 Sequential pallets, Source: Author

3.2.3 Diverging palette

We can use a diverging palette for values with a central value, for example, zero, and numerical ranges from both sides, like -5 to 5. In that case, we would use a combination of two sequential pallets that have the same point (zero). For this type, two different colors are used, one for each side (Figure 3.5). When utilizing a diverging palette, one should pick colors that go with each other. However, colors must be distinctive because otherwise, it would be hard to represent the lowest and highest values. (Yi, 2019)

Figure 3.5 Diverging palette, Source: Author

Apart from the types above, two others should be mentioned, **discrete** and **continuous** palette (Figure 3.6).

A discrete set of colors means that every range of numbers gets its own shade. In contrast, a continuous palette blends further with the numbers. Using a discrete palette is recommended as it is easier to represent differences. (Yi, 2019)

Figure 3.6 Discrete vs continuous palette, Source: (Yi, 2019)

3.3 Font and text

So far, the term 'dashboard' has been used in connection to visuals or graphics. As such, the assumption forms that perfect data visualization is simply colorful pictures and charts. This, however, is not correct. If we want to present information that will be comprehendible to all, text should be considered as well. When presented with only a visual, humans may interpret data in different ways. As a result, a group of people might reach varying conclusions about what the data offers

By including text, however, people can arrive at the same conclusions. Still, text needs to pass through many unofficial rules in order to ensure comprehensible and valuable information. For this case, we can use the golden rule: less is more. Some of the advice given by Yuk (Yuk et al., 2007) in the book Data Visualization for Dummies is:

- **Few words:** Using only a few words to elevate visuals is crucial for many reasons. There is not much space, so adding too much text would make everything crowded and hard to read. It also takes more time to read or can be easily misinterpreted

- **Simplicity is a key:** It is advised to use a simple word, a single line of text, or even a few characters because there is a low risk of misinterpreting. The reader can immediately understand visuals if we turn to simplicity. One of the examples can be using "good" instead of "positive."

- **Colors:** It is recommended to use only a few neutral colors to present text; otherwise, it can contrast the visuals and look messy. Also, one should choose a color that matches other colors on the dashboard. Bright colors should not be used for text representation. RAG colors: red, amber and green are widely used for visuals, representing something bad, and highlighting positive and negative changes.

- **Within the range:** All text used should be close to the visual it represents; otherwise can be misunderstood as well as hard to find and read. The text should always be placed horizontally and never vertically.

- **Labeling:** To achieve the best visuals, it is good to use all three text labels: title, description, and value. Each of them has its spot and meaning within the visual. Their size also plays a role, so it is important to make the more essential parts larger, e.g., the title should be the largest.

- **Fonts:** best fonts to use are Times New Roman, Arial, Garamond, and Verdana. It is desirable to use only one of them for the whole data visualization piece.

3.4 Position on a page

When telling a story to a user, we can always single out essential parts that we want other people to see. If there is something vital to show on the dashboard, it should be put it in the page's top-left corner. People tend to read in a zig-zag motion (Figure 3.7), so we should put visuals accordingly by importance. Information that is not that important should end up in the bottom right corner. If we put it unnaturally, it can take time for the reader to find the important things he wants to see. (Knaflic, 2015)

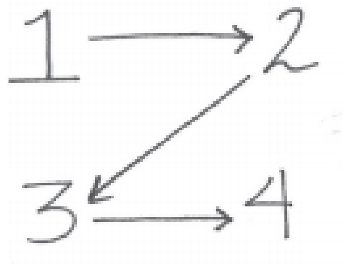


Figure 3.7 Position on a page, Source: (Knaflic, 2015)

3.5 Charts and Graphs Theory

Staple components of data visualization and dashboards are charts and graphs. However, many people do not understand that they should not use all data for all visual types. Each visual has its pros and cons, and in this subchapter, and author explains them in a greater detail using knowledge gained in school and work. We can divide them into groups: time and category charts.

3.5.1 Time charts

It represents data and changes throughout time. Hence there must always be a column with time (month, year, period..).

Line chart
Line charts are suitable for representing trend lines. We use it to show key performance indicators and their targets. It is easy to read values and make comparisons when using this type of chart. Characteristics (Table 3.1) and example (Figure 3.8):

Table 3.1 Line chart, Source: Author

Pros:	Easy to understand and show basic trend lines
Cons:	Unable to see turning point when line is smooth

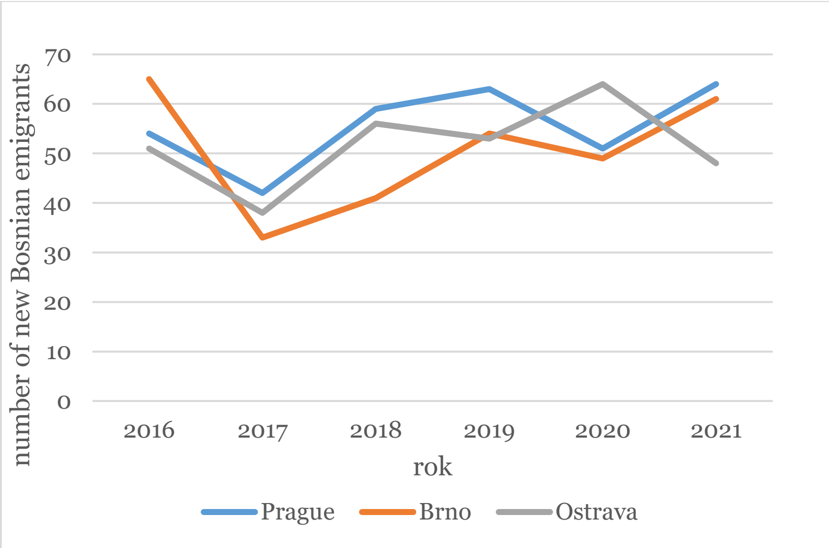


Figure 3.8 Time chart, Source: Author

Column chart
It is one of the most popular and used charts. They have the same function as line charts, but instead of using lines, it uses columns that we can compare next to each other. It can be easier to understand because values can be shown for each column, making reading values more efficient. Characterictsics (Table 3.2) and examples (Figure 3.10) and (Figure 3.9):

Table 3.2 Column chart, Source: Author

Pros:	represents KPIs and their values over time
Cons:	hard to read when too many columns are shown

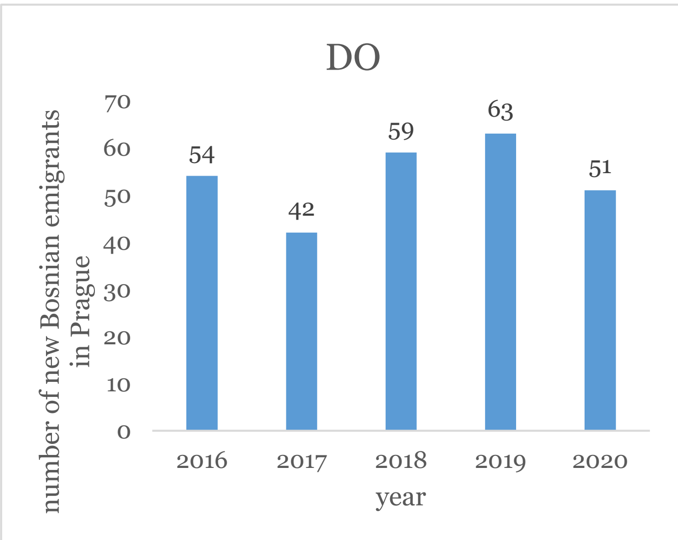


Figure 3.10 Column chart- DO, Source: Author

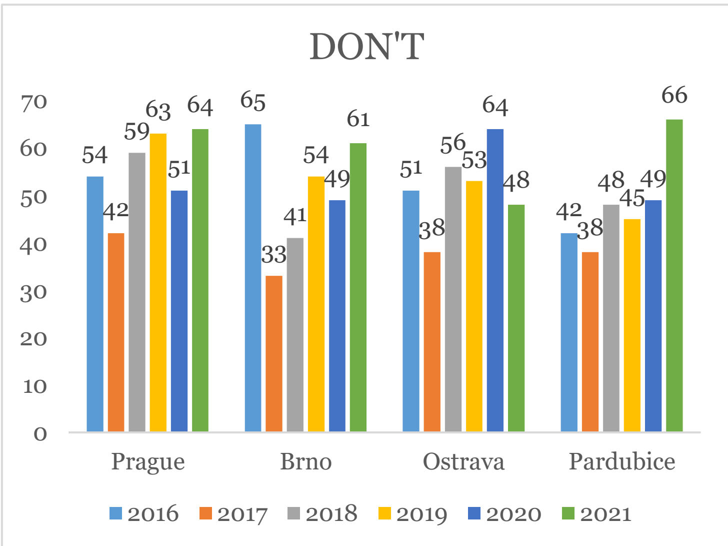


Figure 3.9 Column chart- DON'T, Source: Author

Multiple column chart, with and without line
If we need to compare more KPIs at once, this chart is the best option. However, it may not be the best option for comparing more than three KPIs because it gets difficult to read. To see the set target and movement, we can easily add lines and make the comparison more complex. Characteristics (Table 3.3) and example (Figure 3.11):

Table 3.3 Multiple column chart, Source: Author

Pros:	good for comparison for 1-3 KPIs, able to incorporate lines
Cons:	It looks messy if the number of KPIs goes up

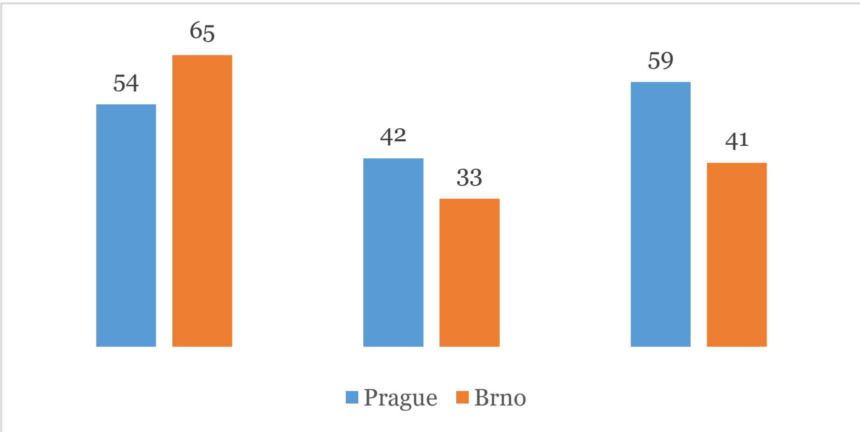


Figure 3.11 Multiple column chart, Source: Author

Area chart
We use it to make trends more prominent to help for future planning.
We can easily spot trends using this type, and it shows more than a typical line chart. Characteristics (Table 3.4) and example (Figure 3.12):

Table 3.4 Area chart, Source: Author

Pros:	shows KPI's target
Cons:	can look wrong and sometimes unnecessary, hard to distingush

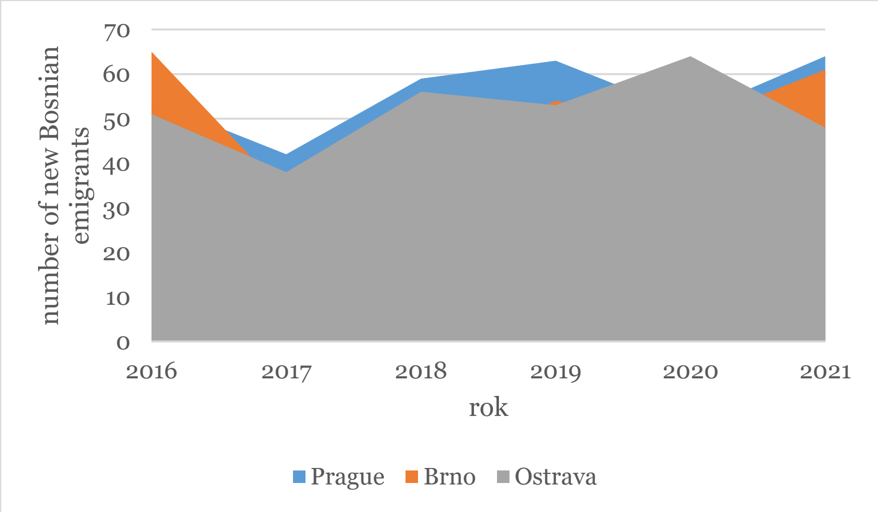


Figure 3.12 Area Chart, Source: Author

Stacked Column

It is most helpful when we have a sum of KPIs but want to see each one's share. Nevertheless, it is rather useless if we need to see the individual trends forming. Characteristics (Table 3.5) and examples (Figure 3.13) and (Figure 3.14):

Table 3.5 Stacked Column, Source: Author

Pros:	see the share of the sum of individual KPI's, overall trend
Cons:	cannot see individual trends nor targets, too complex if we have more KPIs

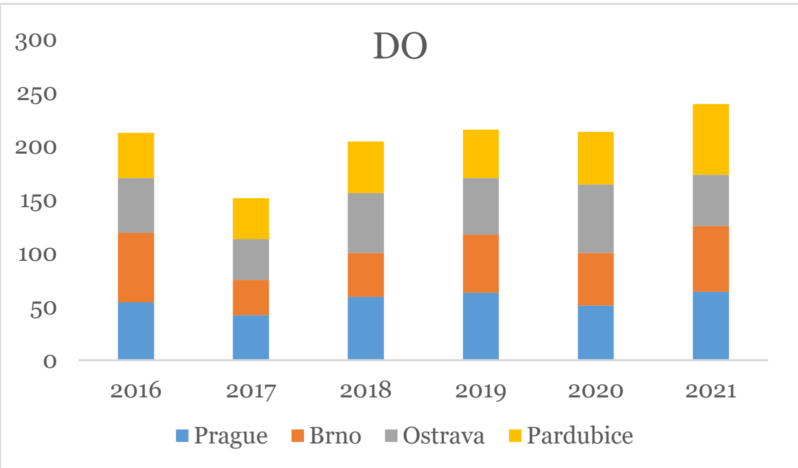


Figure 3.13 Stacked column chart- DO, Source: Author

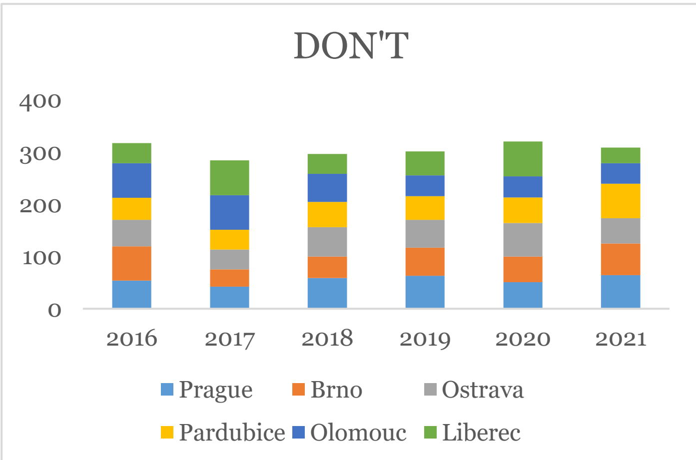


Figure 3.14 Stacked column chart- DON'T, Source: Author

Horizontal bar

It is similar to the column chart, but only reversed. It is better to show values using this chart, but only if we have few columns. Otherwise, it gets cluttered, and values are then difficult to read and understand. Characteristics (Table 3.6) and example (Figure 3.15) and (Figure 3.16):

Table 3.6 Horizontal bar, Source: Author

Pros:	easier when values must be shown
Cons:	cluttered if there are too many columns

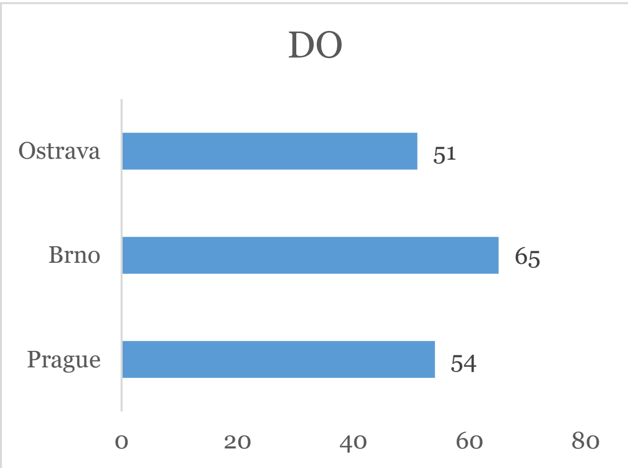


Figure 3.15 Horizontal bar- DO, Source: Author

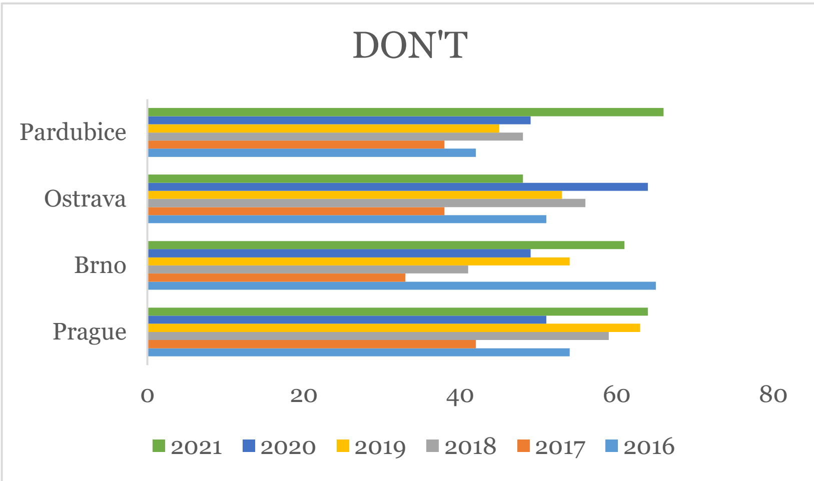


Figure 3.16 Horizontal bar- DON'T, Source: Author

3.5.2 Category charts:

As the name itself suggests, category charts are designed to show a split of different categories. There are time values, but these are not shown in the charts' output; for example, we can use a pie chart and set the date range for the last half a year and offer only categories in that time range. They are not designed to show trends or targets but primarily represent shares of each class of category.

Pie chart

It is ideal if one needs to see percent splits for a specified time. It can be suitable for regions, products, etc. For a better understanding of data, it is better to turn the legend on. Almost the same as pie chart is doughnut chart. Characteristics (Table 3.7) and example (Figure 3.17):

Table 3.7 Pie chart, Source: Author

Pros:	easy to compare categories for a specific period of time
Cons:	one needs to make more pie charts to cover the whole time

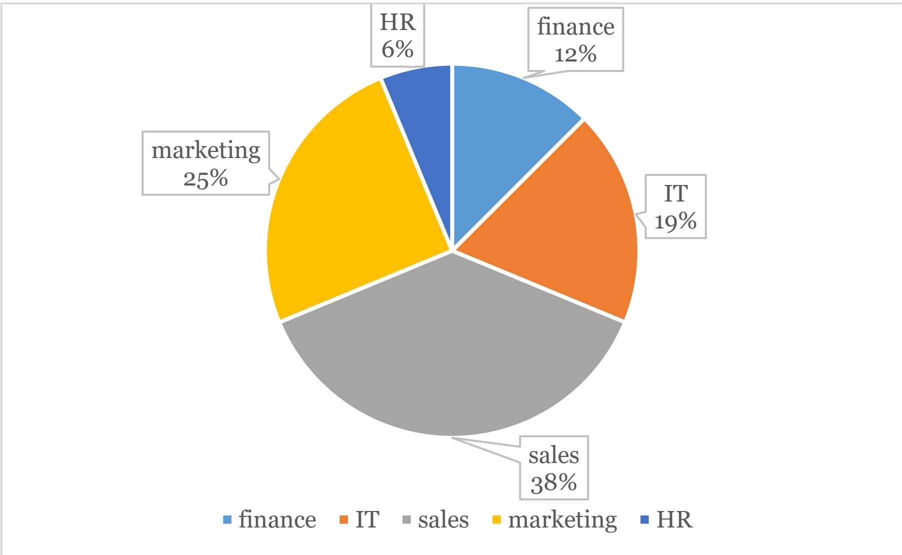


Figure 3.17 Pie Chart, Source: Author

4 Introduction to the Embassy of Bosnia and Herzegovina

This chapter aims to introduce the reader to the work and functions of Bosnia and Herzegovina's embassies, especially the one based in the Czech Republic that will be later used as the leading example.

To begin this chapter, we need to define what an embassy is, and Collins dictionary defines it like this: "*An embassy is a group of government officials, headed by an ambassador, who represent their government in a foreign country. The building in which they work is also called an embassy.*" (Collins, n.d.)

Bosnia and Herzegovina and its foreign policy

Bosnia and Herzegovina is a relatively young country that gained its independence at the beginning of nineties. Not long after gaining independence, Bosnia became part of a war that lasted for four years between 1991 and 1995. It is estimated that around 1.5 to 2 million people left Bosnia during that time, which is more than one-third of the Bosnian population. The population of Bosnia and Herzegovina now estimates to be around three and a half million people. One of Bosnia's leading diplomacy goals was to establish consular missions in the countries with many Bosnian war immigrants.

Activities of BiH foreign policy have two characters: bilateral and multilateral.
Bilateralism or bilateral relations are relations between two countries and can be either political, economic, or cultural. The difference with **multilateralism** is that it is a conduct of relations between multiple countries that have the same goal.

Some of the priorities of BiH foreign policy are the protection of independence and territorial integrity, implementation of General Peace Agreement (GPA), inclusion in integration processes, and participation in bilateral and multilateral activities with systems like UN OSCE, OIC and others. (Paravac, 2003).
To its priorities, we must include the protection of Bosnian and Herzegovinian citizens' interests in foreign countries. This is especially the case in bilateral activities, where there is a goal of protecting and assisting as many BiH citizens in need.

Pillars of foreign policy

BiH's foreign policy pillars should not be viewed as fixed points, but rather dynamical guidelines towards creating and preserving active policies toward EU, NATO, the region, global issues, etc. They are the strategic directions based on principles, international agreements, and some of the priorities mentioned above. Based on global, local, and regional changes and the position of BiH as a small country in Europe, some of the pillars are (Čović, 2018):

- security and stability.
- economic prosperity.
- protection of the interest of BIH citizens abroad and international legal cooperation.
- promotion in the world.

To better understand why embassies need to have an active list of nationals in different countries, it is important to include how and what kind of help the embassy gives to its people. Some are (MVP BIH, n.d):

- lost or theft of documents, in which case the consular department can either issue a travel document to return to Bosnia or issue a new passport, depending on conditions
- in case of arrest, the embassy can assist in cases like this by calling the local organs and giving the consular protection and analysis of the case
- illness, in which case the person will be advised about health facilities
- death, in which case the consular employee would notify the family and help with law formalities
- various problems

Embassy of Bosnia and Herzegovina in Prague, Czech Republic

One of the countries that helped many Bosnians immigrate during the war is the Czech Republic, where there are around 10 000 Bosnian emigrants and war refugees. The Czech Republic recognized BiH as an independent state in 1993. In the first years of immigrating to the Czech Republic, Bosnians did not have any source of help from their state, so the only aid was given by Czech people, who assisted them them with visas, procedures, jobs, and schools. One such person was Dana Němcová, who got nominated for the medal of The Presidency of BIH for her impressive work, in fact many Bosnians in the Czech Republic call her their mother. (Oslobođenje, 2020)

The Embassy in Vienna provided the first consular help, but as the needs grew more prominent, it was decided to open the embassy in Czech Republic, which also covers the Republic of Slovakia. The Embassy was opened in 2005, and in addition to consular affairs, it offers a range of support, such as assistance in obtaining passports and improving bilateral relations. (BH embassy, n.d.). The head is the ambassador, counselor, consular department with the consular department head and consular employee.

In the Czech Republic, few organizations of ex-YU people organize many activities and events with the Embassy, creating a need to have a list of people in the above countries. Also, during the elections in Bosnia and Hercegovina, it is essential to have information about Bosnian emigrants. Unfortunately, the BiH embassy does not keep record of such information.

Current situation:

Similar to the country, the BiH embassy in the Czech Republic is still developing while moving further with constant planning and many strategies. The main problem is the lack of digitalization. There are no used systems for records of BiH nationals abroad in the current situation. Consequently, most of the essential information is kept on papers or poor excel tables. Data found on papers or excel tables are primarily incomplete and do not fulfill the requirements defined in the next chapter. Because of this, there is no adequate information about Bosnian emigrants living in the Czech Republic. The central system for any work is Office 365 because it is easily accessible, free, and available for everyone and every computer.
Facebook is used to share important information with people, but that is not enough since not everyone sees these messages.

Needs of the Embassy of BiH in Prague

The embassy of BiH in Prague has many needs. However, some are more important and crucial to be fulfilled, so the embassy can develop further and build better relationships with emigrants. They want to be able to send out information to particular groups of people that relate to it.

- Digitalization of embassy's environment.

The primary need is to digitize some processes, like the registration of emigrants and their data analysis. This would be beneficial for all sides, thus this project should be taken a step further. It would allow growth and a chance to be more technically equipped.

- Having a SW application for registration of Bosnian emigrants

To gain data and get information, it is mandatory to have a functioning application to help employees register people. It provides a lower risk of having non-validated data, as an interactive application already validates data, but users of the app confirm it.

- To have all information and situation view of emigrants in one place.

It is imperative to have everything organized in one place since it simplifies processes, and there is no need to search all papers and incomplete tables. This is also connected with the need to have a view of the emigrant's situation in Prague. Thus, it is essential to have a dashboard that would give an insight into things and show opportunities or risks that the embassy could later influence.

5 SW application for solving the registration of emigrants

This chapter is the beginning of the practical part of this bachelor thesis. It is the documentation for the creation of the SW application for the registration of Bosnian emigrants. It contains two parts: the design of the form and the main table's design, which contains data from the form. It provides a view into formulas and VBA code used for the creation of both parts. Since the primary problem is the lack of digitalization, it is crucial to lead the embassy towards it. Moreover, the first step would be implementing a system that is easily accessible. The author used the agile approach for every step of the SW application life cycle to make sure to set up the right goals, needs and to define requirements. Using this approach, the author was able to create the application and dashboard much faster than without regular consultations. Moreover, the author could be assured that the development is going in the right and expected direction, ensuring that the needs and requirements are fulfilled expectedly.

5.1 The need for the SW application and database

To get insight into the current situation and the needs in the Embassy of BiH in Prague, the author used the fact that a family member is a diplomat in the above-mentioned embassy. This helped to establish communication with other employees, who listed the current problems and shortcomings. The author was also able to see first-hand how data is poorly stored and how some processes took much time.

The idea to create the SW solution came up from the needs of employees of the Embassy of BiH in Prague. State institutions in Bosnia are planning a project of obtaining yearly reports about the diaspora structure from the embassies. This requires the Embassy to have a reporting system, which will be discussed in the next chapter. The Embassy should dispose of data used as the source for analysis and implement such a system. Nevertheless, the biggest problem in their environment is the lack of digitalization and not having a system for the evidence of Bosnian emigrants in the Czech Republic. Most of the information is stored in poorly created Excel tables or papers and is usually incomplete. Therefore, it cannot be used as the data source for the dashboard. The SW application and database would not only benefit the employees who work with such information but also the whole Embassy because all data would be stored in one place, and registration tasks would be quickly done. Moreover, it will be used as a source for the reports. For example, if an employee needs to send information to a specific group of people, it would take much time to go through all data stored in papers or excel documents to find certain information.

Another example could be a user trying to add a new person to the table. Typing everything by letter can take a long time, and also creates a risk of making mistakes that users are not aware of. Each of these processes could take around 5 minutes, but the goal is to optimize them and make them more effective, where it will take only a few seconds to do the task. Another reason for such an SW solution is the potential of the digitalized world as such, which, unfortunately, the Embassy is not yet part of. Using such an application would be a step forward to digitalization. More details on the needs and current situation can be found at the end of the previous chapter that is about the Embassy itself.

5.2 Description of application

The purpose of the application is to provide the Embassy's employees with more powerful ways of working. Using the registration form, database, and many filters, end-users can work faster and more effectively. Application is built in Excel and VBA and should be used on updated OS versions with Office 365. It is built in Excel because it is for free and accessible by everyone. Especially it is an excellent first step towards digitalization that the Embassy lacks. Some of the machines may have to be adapted to get the full functionality of this SW solution. Basic functionality is creating new records using the registration form. Data is stored in a database and can be edited, deleted, and manipulated using filters.

5.3 Requirements for the application and database table

This subchapter defines the fundamental requirements and functionality for developing the SW application and database table. One of the essential parts of development occurs at this stage. It is the guide that will help the developer understand how the end product should look and work. Every requirement was defined in a meeting between the employees and the author to better understand the employees' concerns and needs. In the case of usage, some parts of the application will be more defined, like security passwords. The most important condition was the user-friendly interface since not all of the employees are technically oriented.

The most significant role was the overall functionality, such as quickly creating new records, properly storing the data, and manipulating them. When creating a new record, the end-user should not worry about creating an ID number; thus, it was essential that every new empty and unfilled form already have a generated ID number. To ensure that all data is filled in correctly, it was suggested to use data validation controls and controls that check if mandatory polls are filled in before saving the new record. Saving the new record should be quick and easy and should be continuously added to the top part of the database table on another worksheet. When it comes to working with the database table, the most significant request was filtering system that would help employees generate the data of the specific groups.

Apart from filtering, it was mandatory to ensure that every record can be either updated or deleted. One of the requirements that the author suggested is next-level security. No one else must have access in addition to the agreed and authorized users. Worksheets containing registration forms and databases are very well hidden and can be accessed only by password.

More about functionalities will be defined in the following subchapters. More detailed requirements for the application can be found in the table (Table 5.1), and the requirements for the database table are presented in the table (Table 5.2).

Table 5.1 Requirements for the registration form, Source: Author

	Requirements for the application (form)
1	The application was made to enable a more manageable and digital way of registering emigrants from Bosnia and Herzegovina.
2	It should be user-friendly with minimal effort from employees.
3	The benefit of the form is straightforward entry to the main table.
4	Not all data entries are mandatory, and the form should control this problem before saving the entry to the table. If missing mandatory entries, it should not be possible to save the data to the table.
5	When inputting the data, every cell should have data validation control for either format, length, or lists.
6	It should have the functionality of creating an empty form with an already generated ID number.
7	The form must be well secured from more points of view. It must be protected from the user itself, meaning that he/she should not be able to make changes to the form.
8	Security of the form must be on a higher level. When opening a document, it must be very well hidden, so no one can even unhide it. To reveal the sheet with the form, the user must input the password into to entry box. When saving and closing the document, the mentioned sheet should disappear again.

Table 5.2 Requirement for the table of records, Source: Author

	Requirements for the table of records
1	The table should be able to receive data entries from the form.
2	Users should be able to edit and delete records and save changes
3	Filters should be by last name, nationality, citizenship, residence and info, and should be also used as combined. When finished with filtering, the user should be able to get back to all data.
4	Security of the database table must be on a higher level. When opening a document, it must be very well hidden, so no one can even unhide it. To reveal the sheet with the form, the user must input the password into to entry box. When saving and closing the document, the mentioned sheet should disappear again.
5	Table must contain categories that were either defined or created based on requests from the end-users

5.3.1 Users

Every employee of the Embassy will be able or at some point required to use the application and dashboard. Therefore, there was no need to establish different security options and access rights for different types of users. They will share one unique password to enter the database and form. Every time a diplomat comes or leaves the Embassy, the password will be changed. More about particular types of employees was discussed in the previous chapter.

5.4 Design and implementation of the form

5.4.1 Graphic design of the form

The data entry form should be user-friendly and graphically good-looking. Nevertheless, it should contain text boxes for data input and buttons to work with it. When creating the form, it was essential to decide which way to make it. Two versions were made, the first (see Figure 5.2) was made using the Form tool in VBA, and a second one (see Figure 5.1) (Annex 2) was created on the sheet having unlocked cells as text boxes. Content and functionality were on the similar level for both forms, but the knowledge of creating and its possibilities in VBA was lacking, so the form on the sheet seemed like the better choice. The most significant role in choosing the right one were the graphics. The VBA form does not look good as the sheets form, so it was not a convenient option to choose. Based on the above requirements, the chosen form was the sheet one.

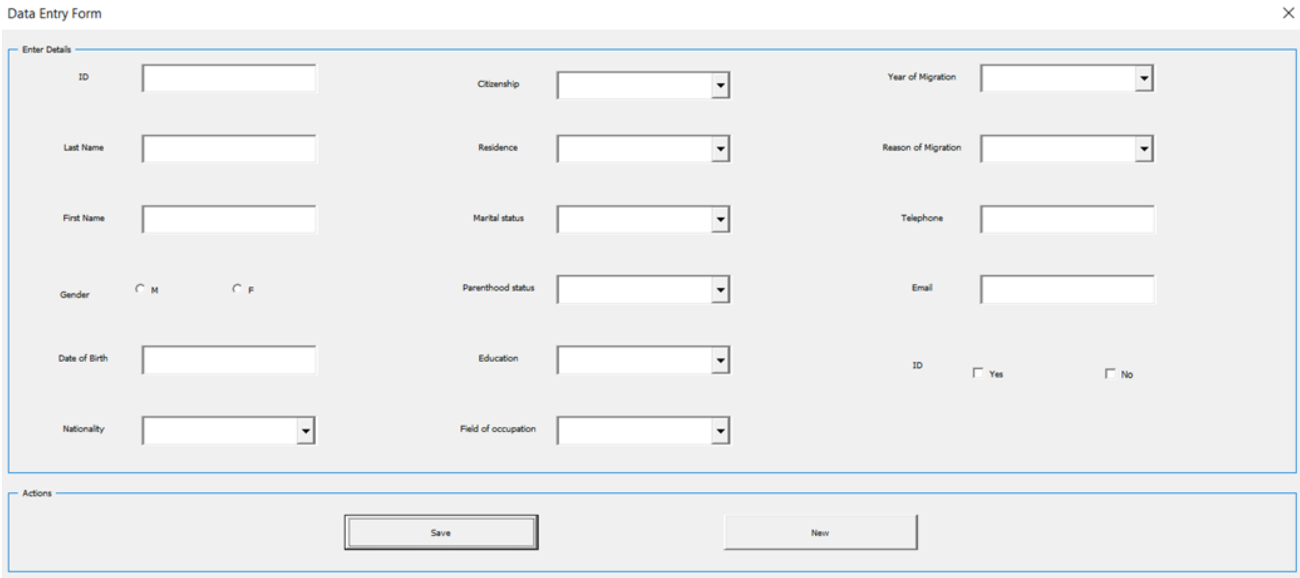


Figure 5.2 User form made in VBA, Source: Author

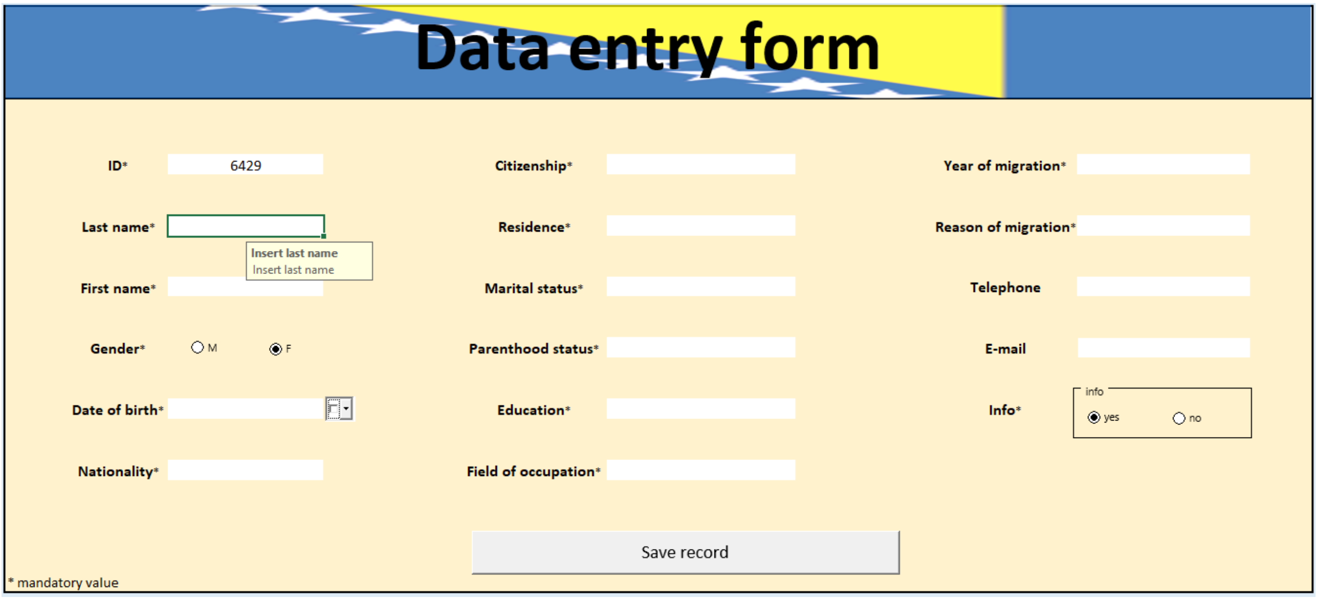


Figure 5.1 User form made in a worksheet, Source: Author

5.4.2 Data entries and their validation

The form has seventeen data entry boxes conditioned by the side of embassy employees, stating that all of them are important and should be used individually. The title part has the Bosnian flag as a representation and better graphics. Some information is mandatory, which is represented by star sign *. All of them have data validation controls based on either format, list, or length. They also have messages with short instructions and error pop-up boxes if the entry is against the rules. Some of the longer lists' values are sorted alphabetically or by number, so it is easier to choose. The worksheet is locked, except for these cells, so the user cannot make any other changes except the defined cells. This also allows the user to move through the cells by using Enter or down.

Every mandatory cell has a VBA code behind the "Save record" button, controlling if it is filled in. More about it will be in the subchapter about the buttons used in a form.

To create some of the data validations and functionality of some cells, it was essential to create a side worksheet called "help". In the output, it remains hidden.

ID (Long, mandatory)

It creates a new ID number for the person. It is locked, so users do not have to worry about it or accidentally destroy it. It is mandatory, but since it is locked in, it will never be empty. A click on the button "Save record" will automatically generate a new ordinal number for the next record, after saving previous to the table. This works by creating a reference of the last input ID row in the table and adding it up by one.

Last name (String, mandatory)

This cell is used for the input of the person's last name. Data validation was created by the unique format that allows only letters, space, and sign -. By allowing space and sign -, the user does not have a problem with the entry if the emigrant has multiple last names. If typed in incorrectly, a message box pops up, and the user must try again. The data validation formula is (Figure 5.3):

```
=KDYŽ(D11="",PRAVDA,KDYŽ(JE.CHYBHODN(SOUČIN.SKALÁRNÍ(HLEDAT(ČÁST(D11,ŘÁDEK(NEPŘÍMÝ.ODKAZ("1:"&DÉLKA(D11))),1),"abcdefghijklmnoprstuvzqwyxčšž- "))),NEPRAVDA,PRAVDA))
```

Figure 5.3 Last name validation formula, Source: Author

First name (String, mandatory)

This cell is used for the input of the person's first name. Data validation was created by the unique format that allows only letters, space, and sign -. By allowing space and sign -, the user does not have a problem with the entry if the emigrant has multiple first names. If typed in incorrectly, a message box pops up, and the user must try again. Data validation formula is same as the above presented one for the last name entry (Figure 5.3).

Gender (String, mandatory)

For the creation, a switch was used. It does not contain any data validation since it is impossible to leave it empty or choose something else. User can choose between M as male, and F as female.

Date of birth (Date, mandatory)

Input can go both ways, either choosing from the calendar or typing in the date. The format is month/day/year. It has data validation that the chosen date must be between 1/1/1900 and 12/31/2100. The calendar (Figure 5.4) was made using a date picker, and unfortunately, it does not work if a person does not have it added to their Excel. Hence, it is advised to install it.

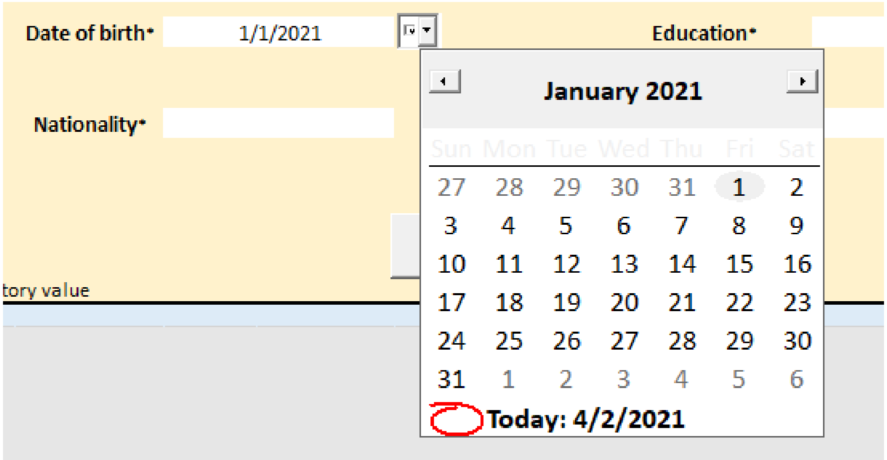


Figure 5.4 Date picker calendar, Source: Author

Nationality (String, mandatory)

This entry contains only list values: Bosniak, Croatian, Serbian, or other. Data validation was set up based on this list and nothing else can be inserted.

Citizenship (String, mandatory)

This entry contains only list values: BIH, CZ, or Double. Data validation was set up based on this list and nothing else can be inserted.

Residence (String, mandatory)

This entry contains only list values that are regions: Hlavní město Praha, Jihočeský kraj, Jihomoravský kraj, Karlovarský kraj, Vysočina, Královéhradecký kraj, Liberecký kraj, Moravskoslezský kraj, Olomoucký kraj, Pardubický kraj, Plzeňský kraj, Pardubický kraj, Středočeský kraj, Ústecký kraj, Zlínský kraj. Data validation was set up based on this list and nothing else can be inserted. These entries are written in Czech language, for the proper function of the map chart in the dashboard.

Marital status (String, mandatory)

This entry contains only list values: single, divorced, in a relationship, married, or widowed. Data validation was set up based on this list and nothing else can be inserted.

Parenthood status (String, mandatory)

This entry contains only list values: no kids, kids, or single parent. Data validation was set up based on this list and nothing else can be inserted.

Education (String, mandatory)

This entry contains only list values: university degree, secondary education, or primary education. Data validation was set up based on this list and nothing else can be inserted.

Field of occupation (String, mandatory)

This entry contains only list values: Arts/Entertainment, business, education, environment, gastronomy, government, healthcare, law&public policy, sciences, social impact, technology, tourism or other. Data validation was set up based on this list and nothing else can be inserted.

Year of migration (String, mandatory)

Shows the range of 10 years between the 1990s (war period) and today. This entry contains only list values: 1990-2000, 2001- 2010, 2011- 2020 or 2021-. Data validation was set up based on this list and nothing else can be inserted. Reason for not using the single year is that the employees stated that they do not need it and they preferred the version of having few periods to choose.

Reason of migration (String, mandatory)

This entry contains only list values: war, family, work, other or studies. Data validation was set up based on this list and nothing else can be inserted.

Telephone (Variant, not mandatory)

A unique format for phone numbers was used to validate this cell. For example, it is enough to write the numbers when inputting, and cell automatically converts it to a defined format. Data validation formula (Figure 5.5):

```
=KDYŽ(N14="",PRAVDA,KDYŽ(JE.CHYBHODN(SOUČIN.SKALÁRNÍ(HLEDAT(ČÁST(N14,ŘÁDEK(NEPŘÍMÝ.ODKAZ("1:"&DÉLKA(N14))),1),"0123456789+"))),NEPRAVDA,PRAVDA))
```

Figure 5.5 Telephone number validation formula, Source: Author

Email (Variant, not mandatory)

To input data in this cell, a specific format must be respected. Data validation was set up to check if @ was used, if there is text before and after it, and in the text after, there must be at least one point ".". There was no need for control of email domains, since many Bosnian users have many different ones. Data validation formula (Figure 5.6):

```
=JE.ČISLO(POZVYHLEDAT("*@*.?*",N17,0))
```

Figure 5.6 E-mail validation formula, Source: Author

Info (String, mandatory)

For the creation, a switch was used. It does not contain any data validation since it is impossible to leave it empty or choose something else. User can choose between yes or no values, based on emigrant's decision about receiving information from the Embassy.

5.4.3 Button "Save record"

To make the data entry form functional, it was crucial to import button with macros behind it. The basic functionality of the form is ensured by creating a button "Save Record". It was made using macros in Excel and later edited in VBA code, by adding field controls and generating new ID number.

The button "Save record" has an extended code behind it. To be able to save a new record to the Database worksheet, it was crucial to create another worksheet that holds the references of cells from the "Form" worksheet. The "Save record" button works in the way that it copies the whole row from the help worksheet, adds a new row at the top of the database table, and pastes the record there.

Behind the "Save record" button is also a validation that all mandatory entries are filled in, and if so, then a message box pops up informing about success, and the process of saving goes through. Otherwise, another message box pops up informing about the mistake, for example: "Last name not filled in", and the record will not be saved until everything is correctly filled in. Code for the validation requires that the mandatory fields are filled in before saving a new record (Figure 5.7). One of the codes for the validation itself (Figure 5.8) can be found below. Other codes were created in a same way as the "Kontrola_lastname()".

```
Public Sub KontolaZapis()
If Not Kontrola_lastname() Then Exit Sub
If Not Kontrola_firstname() Then Exit Sub
If Not Kontrola_dateofbirth() Then Exit Sub
If Not Kontrola_nationality() Then Exit Sub
If Not Kontrola_citizenship() Then Exit Sub
If Not Kontrola_residence() Then Exit Sub
If Not Kontrola_mstatus() Then Exit Sub
If Not Kontrola_pstatus() Then Exit Sub
If Not Kontrola_education() Then Exit Sub
If Not Kontrola_fieldofo() Then Exit Sub
If Not Kontrola_yearofmig() Then Exit Sub
If Not Kontrola_reasonofmig() Then Exit Sub

tlacitko_save
tlacitko_new

End Sub
```

Figure 5.7 Code for controls of mandatory entries, Source: Author

```
Private Function Kontrola_lastname()
Dim lastname As String
lastname = Sheets("DoNotTouch").Range("B2").Value
If lastname = "0" Then
Sheets("Form").Range("D11:E11").Select
MsgBox ("Mistake- last name must be filled in")
Kontrola_lastname = False
Else
Kontrola_lastname = True
End If
End Function
```

Figure 5.8 VBA code if last name is filled in, Source: Author

Apart from other functionalities, when clicking on this button, all data entry cells will be cleared out from the previous record and new ID number is generated, so the form is then ready to use. When generating a new ID number, it searches the database table, locates the last ID number created, and adds it up by one. That way, users do not have to worry about it. This worksheet is always extra protected with the lock function, and only data entry cells are unlocked and able to use. With the use of the button, the sheet is unlocked, used, and locked again. Without unlocking the sheets, it is impossible to save data in the table, thus it must be firstly unlocked and when finished locked again. Code behind this functionality, author used another sub: (Figure 5.9).

```
Sub tlacitko_new()
'
' tlacitko_new Makro
'
ActiveSheet.Unprotect
'
    Range("D8:E23").Select
    Selection.ClearContents
    Range("I8:J23").Select
    Selection.ClearContents
    Range("N8:O17").Select
    Selection.ClearContents
    Range("D8:E8").Select
    ActiveCell.FormulaR1C1 = "=Database!R[-5]C[-3]"
    Range("D8:E8").Select
    ActiveCell.FormulaR1C1 = "=Database!R[-5]C[-3]+1"  '// part for generating new ID number
    Range("D11:E11").Select                            ' // always positiones to the first open cell: Last name

ActiveSheet.Protect
End Sub
```

Figure 5.9 VBA code for creating new record, Source: Author

5.5 Design and implementation of the table

Since the table is a vital component, it was mandatory to make it correctly and to ensure its functionality by some of the Embassy workers' requirements and requirements from the author's side. Detailed requirements for the table can be found in the previous subchapter.

The table is made of seventeen columns/categories that are incorporated in the form, and they are also explained in the subchapter of Data entries and their validation. Some of the categories were already defined by the Embassy's data, such as last or first name. However, some categories like "year of migration" or "info" were added on their request since there was a need to have it in the person's information. Every row has turned on the same data validation as used in the form. This is to prevent mistakes when updating the rows. To ensure that users do not make mistakes, the worksheet is always locked, except for the text boxes. Filters and buttons work because the author used: “ActiveSheet.Unprotect” at the beginning and “ActiveSheet.Protect” and the end, functions that ensure the functionality of modes even on a locked sheet.

Data used in the table had to be created from scratch because it is forbidden to use the actual data. In order to present and test its functionality, create more possibilities when creating the dashboard, and show users how it would look in real life, it was crucial to create false data for the new categories.

The following picture (Figure 5.10) (Annex 3) is representing the design of the database table. Since it has 17 categories, not all can be seen at once. At the top left, the Search is located by last name filter and the total number of records. The middle contains various filters used for data manipulation. The "Update records” and Delete records" are located at the top right. These buttons and text boxes are used for updating or deleting records. All functions and codes will be explained later in this chapter.

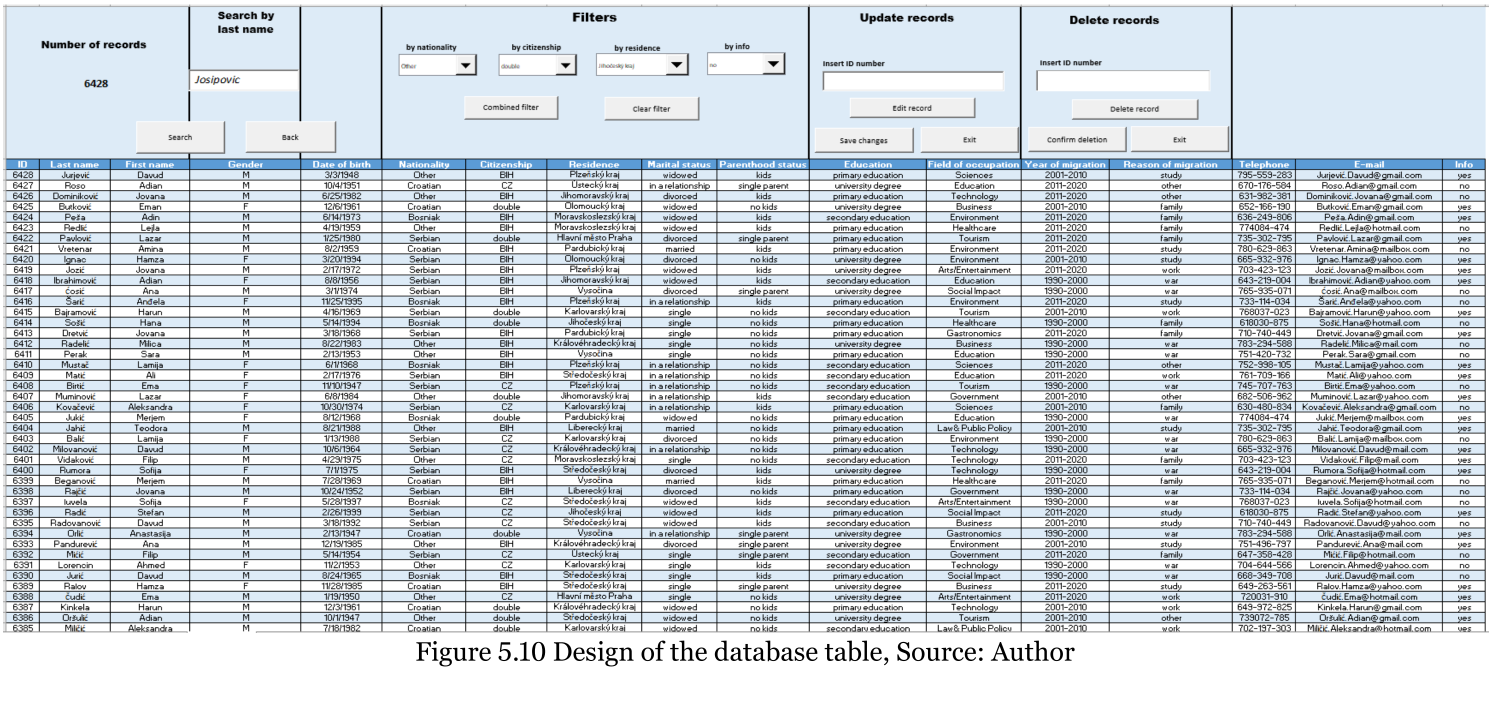


Figure 5.10 Design of the database table, Source: Author

5.5.1 Filters and buttons

Filters are a big part of data manipulation. All that are used in the application were defined in the requirements, and they were created on request by the end-users. Every filter was created using the advanced filtering tool. It is possible to filter data by last name, nationality, citizenship, residence, and info. All four, apart from the last name, can function together as a combined filter. To show back all data, users can use buttons “Back “or “Clear filter”, which turn off filtering mode. Buttons “Search”, “Back”, “Combined filter” and “Clear filter” were created for the easier manipulation of filters.

Filters

Search by last name function

This function filters data and displays the rows that are matching the given string. The user should only enter the desired last name and click on the “Search”. Once the user is done with the search, all data in the table can be shown again by using the “Back” button. If user inserts a last name that is not in the table, the message box will pop up informing about the error. Textbox then gets cleared so user can try again. Code for the “Search” button is presented below (Figure 5.11). Design of this part is presented in the following picture (Figure 5.12)

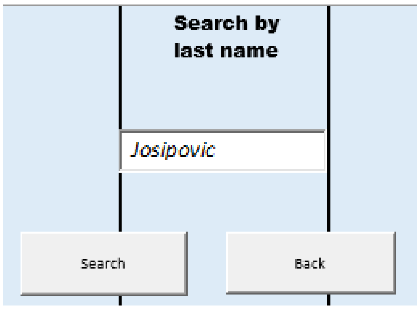


Figure 5.12 Search by last name function, Source: Author

```
Sub Search_by_last_name_fin()
ActiveSheet.Unprotect
Dim ws As Worksheet
Dim wss As Worksheet
Set ws = Worksheets("Database")
Set wss = Worksheets("help")
If Application.WorksheetFunction.CountIf(ws.Range("last_name"), wss.Range("D22")) = 0 Then
MsgBox "Last name not found, try again"
Sheets("Database").TextBox1.Text = ""
ActiveSheet.Protect
Else
Range("last_name").AdvancedFilter Action:=xlFilterInPlace, CriteriaRange:=Sheets( _
        "help").Range("D21:D22"), Unique:=False
    ActiveWindow.ScrollRow = 5791
    ActiveWindow.ScrollRow = 1
    ActiveSheet.Protect
End If
End Sub
```

Figure 5.11 VBA code for filter by last name, Source: Author

Button “Search”:

Its basic functionality is to search the "last name" column and find the cell's matching string. Once it has found it, it shows only the matching ones, making it easy for the users to update or delete them. If there are no matching values, the message box pops up informing about the error and text box gets cleared. Filtering is done by using the advanced filter.

Filters by nationality, citizenship, residence and info

All four filters work on the same principle, using a drop-down list. They can work as individual filters, or they can be combined. When used individually, it is enough to choose from a list, and it immediately filters the data. To use it combined, it is mandatory to use the “Combined filter” button to start the process. Once done, using the “Clear filter” button, all data will be presented again. The design of the Filters tab can be found in a following picture (Figure 5.13). VBA codes used for the individual filter by citizenship is shown in the picture (Figure 5.14) and it represents the advanced filter mode turning on. The deletion of filtering mode is shown in the picture (Figure 5.15).

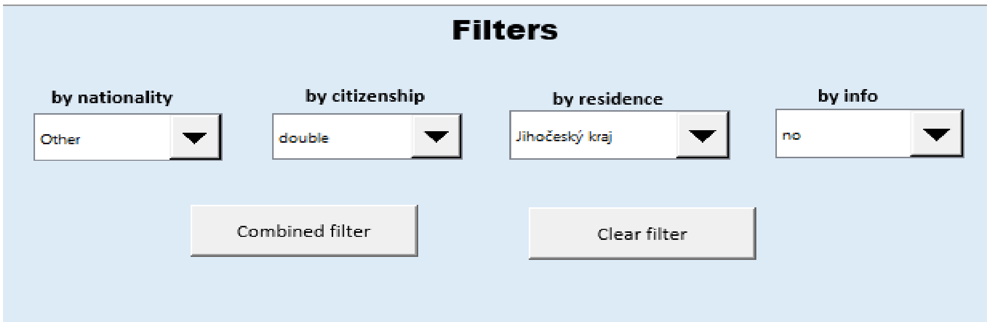


Figure 5.13 Design of filters panel, Source: Author

```
Sub citizenship()
'
' citizenship Makro
'
ActiveSheet.Unprotect

    On Error Resume Next
     ActiveSheet.ShowAllData
     On Error GoTo 0
     Application.CutCopyMode = False
     Application.CutCopyMode = False
     Application.CutCopyMode = False
     'ActiveSheet.ShowAllData
     Range("Citizenship1").AdvancedFilter Action:=xlFilterInPlace, _
         CriteriaRange:=Sheets("help").Range("C13:C14"), Unique:=False
     ActiveWindow.ScrollColumn = 1

     ActiveSheet.Protect
End Sub
```

Figure 5.14 VBA code for filter by citizenship, Source: Author

```
Sub delete_filter()
'
' delete_filter Makro
'
ActiveSheet.Unprotect
'
    ActiveSheet.ShowAllData
    Range("Tabulka7[[#Headers],[ID]]").Select
ActiveSheet.Protect
End Sub
```

Figure 5.15 VBA code to remove the filter, Source: Author

"Combined filter" button:

This button filters data by the choices made in the drop-down lists. It combines the filter of nationality, citizenship, residence, and info. The code used for its creation is shown in the picture below. (Figure 5.16).

```
Sub combined_filter()
'
' combined_filter Makro
'

'
    Application.CutCopyMode = False
    Application.CutCopyMode = False
    Application.CutCopyMode = False
    Range("Tabulka7[#All]").AdvancedFilter Action:=xlFilterInPlace, _
        CriteriaRange:=Sheets("help").Range("F24:I25"), Unique:=False
    ActiveWindow.ScrollColumn = 1
    ActiveWindow.SmallScroll Down:=-20
    Range("B192").Select
End Sub
```

Figure 5.16 VBA code for combined filter, Source: Author

5.6 Edit of records

Every database should have options for inserting, updating, and deleting data stored inside it. Inserting a new record is already explained in the previous subchapters. The ability to make changes or remove rows is not just a requirement of the embassy, but it is a requirement for the correct database operation. Hence, it was mandatory to include it.

The conventional solution for updating the record is taking selected data to the isolated field or form and perform changes that would be saved in the main table. Unfortunately, the author faced difficulties when trying to create this function. Thus, due to problems and the lack of knowledge, another solution was suggested and incorporated into the application. For this purpose, three buttons were created: "Edit record", "Save changes" and "Exit".

5.6.1 Updating

To explain the process, it is first essential to say that an individual filter by ID was not created because none of the employees were interested in it. This solution is based on the assumption that the user will firstly filter data to find the correct row. However, if the user already knows the ID number of a row that should be changed, then he/she can proceed immediately to edit mode. The author uses advanced filters for the performance of these functionalities.

To edit the specific record, one must input the ID number into the text box and click on the button "Edit record" (Figure 5.17), and the entry box pops up. The input box is used for entering the ID of a person whose data should be updated. This filters data further, and the desired row shows up. The row automatically gets unlocked so the person can make changes. Another message box was used to inform the user that the row is ready to be updated. Every category can be changed, but it is limited by the data validation rules that are incorporated in the whole database table. The author controls whether entered ID number exists within a table. If yes, user will be guided to the editing mode with the hidden ID column, so he cannot change it. If entered ID number is not found in the table, message box pops up informing user about it and clears up the textbox, making it ready for another try.

Changes in a row are automatically saved as this is a primary Excel function. The purpose of the "Save changes" button (Figure 5.18) is to turn off the filter mode, show all data in the table, and lock the sheet again. This solution is more trick-based and tested on the assumptions that the Embassy's employees would not try to do anything else except for using the button "Save changes". The testing was positive, and since none of the employees questioned it, this solution was left as a part of the application. On the same principle works the "Exit" button, it is meant to be used before making any changes. Codes used for this solution can be found in the following pictures. Author uses the same principles and codes for update and delete functions.

```
Sub edit_by_id_fin()
ActiveSheet.Unprotect
Dim ws As Worksheet
Dim wss As Worksheet
Set ws = Worksheets("Database")
Set wss = Worksheets("help")
If Application.WorksheetFunction.CountIf(ws.Range("CID"), wss.Range("F31")) = 0 Then
MsgBox "Number not found, try again"
Sheets("Database").TextBox2.Text = ""
ActiveSheet.Protect
Else
Range("CID").AdvancedFilter Action:=xlFilterInPlace, CriteriaRange:=Sheets( _
        "help").Range("F30:F31"), Unique:=False
    ActiveWindow.ScrollRow = 5791
    ActiveWindow.ScrollRow = 1
    Columns("A:A").Select
    Selection.EntireColumn.Hidden = True
    Range("B5876").Select
End If

End Sub
```

Figure 5.17 VBA code to find record for update/deletion, Source: Author

```
Sub save_updates_fin()


ActiveSheet.Unprotect
    Range("B3").Select
    On Error Resume Next
    ActiveSheet.ShowAllData
    On Error GoTo 0

    Columns("A").Hidden = False

    ActiveSheet.Protect
    MsgBox ("Edited row saved!")

    Sheets("Database").TextBox2.Text = ""


End Sub
```

Figure 5.18 VBA code for saving updates, Source: Author

5.6.2 Deleting

The process of deleting works in the same way as the Update records function. But, instead of editing the line, the employee can delete the row using the Right Click/ Remove row option. This was based on the theory that users would know how to remove a row from an Excel table. This theory was tested during the consultations between the employees and the author, and it was confirmed that each person knew how to remove the selected rows. Therefore, there was no need to create an additional Delete function, and both sides agreed on this. Nevertheless, author had major problems with making a one button for deletion, because VBA code would always delete the first row of the whole table, and not the desired one. Button "Delete record" works in the same way as "Edit record". Button "Confirm deletion", works in the similar way as "Save changes". It was made to turn off the filtering mode and show all data when clicked on. Button "Confirm deletion" has another functionality to it, and that it to generate the first free ID number on the "Form" worksheet, to ensure that ID number will be generated even if the first line (highest ID number) of the table was deleted.

5.7 Security of form and database table

One of the most significant roles is the security of the form and table. It should be only visible and used by the people who are justified to use it. Thus, it was mandatory to make good security options so no problems would happen in the future. Requirements for the security solution can be found in the following table (Table 5.3).

Table 5.3 Requirements for the security of the application, Source: Author

	Requirements for the security of the application
users	only users with permission and password can see the form and database table
worksheets	should be very well hidden, without an option to unhide them without the password
password	must be strong and must defer from one another, for the double security

The principle of security is based on the hidden worksheet and secret password. However, they are not just hidden, but very well hidden in the background, so one cannot try to unhide them in the sheets bar and options. The hidden worksheets are called "Form" and "Database", and as seen on the picture (Figure 5.19), they are not only visible on the bar, but also the unhide option is frozen. This secures that it is impossible to unhide them without the correct password. In the version where dashboard is part of the Excel solution, it becomes visible once the database gets visible as well.

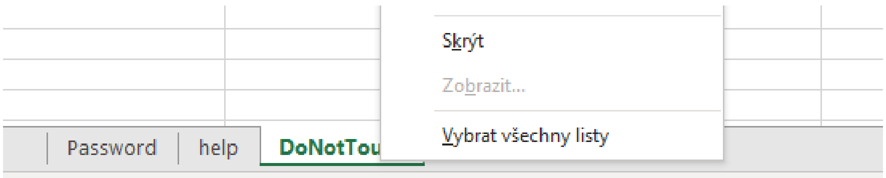


Figure 5.19 Hidden worksheets, Source: Author

When a user opens the Excel file, they see an empty sheet with two buttons (Annex 1). When clicked, the message box will pop up (Figure 5.22), and the user will be able to enter the password. If correct, the desired worksheet will become visible (Figure 5.21). Otherwise, it stays closed (Figure 5.20). The reason for using two different buttons and passwords for two different worksheets is for better security.

Once the user is done working with the file, it is enough to close it, and worksheets will hide again. This way, the process of hiding them is guaranteed.

To enter the form, database and dashboard, the author provided a password in the separate text document, that can be found in the zip files of BT attachments.

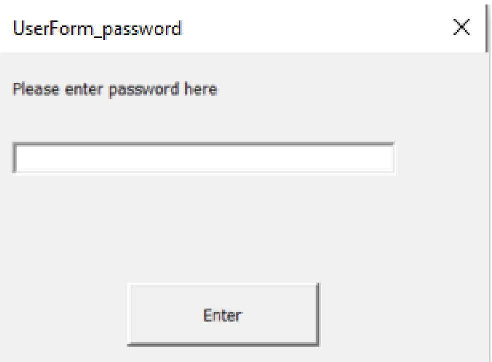


Figure 5.22 Pop up box, Source: Author

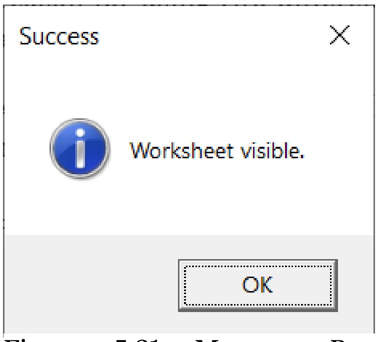


Figure 5.21 Message Box Success, Source: Author

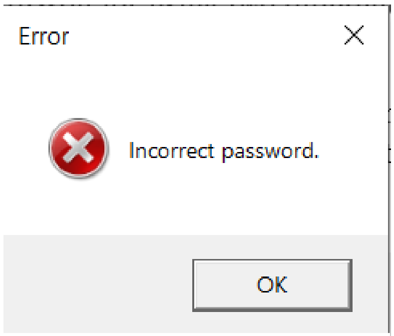
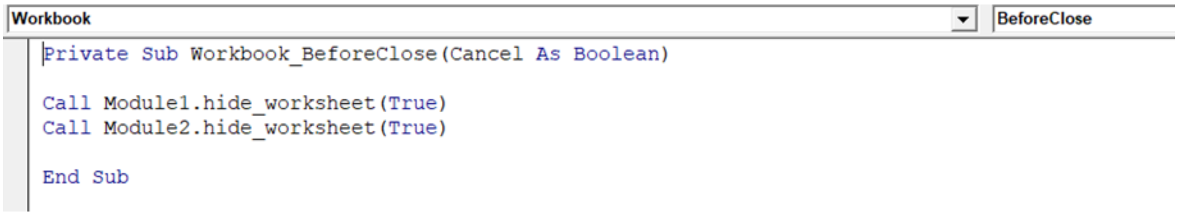


Figure 5.20 Message Box Error, Source: Author




```
Private Sub Workbook_BeforeClose(Cancel As Boolean)

Call Module1.hide_worksheet(True)
Call Module2.hide_worksheet(True)

End Sub
```

Figure 5.23 VBA code for hiding worksheets after closing, Source: Grande, 2016 (edited)

To create this security functionality, it was necessary to create two user forms and two modules, each for one worksheet. Codes can be found below (Figure 5.25) and (Figure 5.24). To ensure that they get hidden every time the document is closed, the author added lines of code to the Workbook code that can be found in the picture above. (Figure 5.23)

```
Sub openUserForm()

UserForm_password.Show
UserForm_password.TextBox1.SetFocus
End Sub
```

```
Sub hide_worksheet(x As Boolean)

If x = True Then

Sheets("Database").Visible = xlVeryHidden

Else

Sheets("Database").Visible = True

End If

End Sub
```

Figure 5.24 VBA code behind user forms, Source: Grande, 2016 (edited)

```
Private Sub enterPassword1()
Dim password As String
password = "excel"

If TextBox1.Value = password Then

Call Module2.hide_worksheet(False)
MsgBox "Worksheet visible.", vbInformation, "Success"

Else

MsgBox "Incorrect password.", vbCritical, "Error"
End If

End Sub
```

```
Private Sub CommandButton1_Click()
Call enterPassword1

End Sub
```

```
Private Sub TextBox1_KeyDown(ByVal KeyCode As MSForms.ReturnInteger, ByVal Shift As Integer)
If KeyCode = 13 Then
Call enterPassword1
End If
End Sub
```

```
Private Sub Label1_Click()

End Sub
```

```
Private Sub UserForm_Click()

End Sub
```

Figure 5.25 VBA code for making worksheets visible, Source: Grande, 2016 (edited)

6 Analysis and design of the dashboard application

In this chapter, the author will introduce the parts of dashboard creation. It contains a description of goals, needs and requirements, defined dimensions and used charts and tables. For this part, the author decided to create two versions of dashboard implementation in two different software. The first version is built in Power BI and has a data model behind it. The second version is made in Excel using regular pivot tables. Reasoning for using Excel was already mentioned in the Chapter 5. The author decided to use Power BI because it is user-friendly and has a great offer of charts and functions. The first part of this chapter contains Introductory study, followed by an explanation of creation in Power BI, and the third part will be dedicated to the dashboard in Excel. Both versions have the same aims and requirements, so the subchapter Introductory study refers to the both solutions. Every step of the dashboard solution was discussed with the employees of the Bosnian Embassy in Prague.

State institutions in Bosnia and Herzegovina are planning for the future years to start the project of following the structure of the Bosnian diaspora around the world. This means that the embassies should report yearly about the changes in the structure.

Nevertheless, it is up to embassies to solve the reporting system. Since the author is connected to the Bosnian Embassy through a family member working there, it was suggested that the author could propose the reporting system. As said above in the thesis, both the application and dashboard can be implemented to any Bosnian Embassy worldwide if they decide to use the offered solutions. The necessity would be to tailor some categories by the needs and create another data source.

6.1 Introductory study

The introductory study contains the explanation of the main goal and requirements, data sources, and analyses the actual condition of the technical equipment. In this part, the author clarifies the describes the process of implementation.

6.1.1 Goals, needs and requirements

To set the precise goal, the author reached out to the representatives of the Embassy. Discussion took place at their main office in Prague. This way, the author could also examine the HW and SW equipment. It was necessary to make sure that OS versions could support software like Power BI. Staff explained that a dashboard as a reporting system is becoming a necessary component of their job. They would need to report the diaspora structure analysis to the state institutions in Bosnia and Herzegovina.

The most significant need was to represent the general structure view of the emigrants in the Czech Republic. They want to know the structure of people in Czechia by regions, gender, nationality, and their interest in receiving information.

Another request was to get insight into the migration of people during different periods, such as the war period. Besides the period of migration, employees stated that they would want to see the main reason for migration to Czechia. Apart from these two main requirements, the author and the staff discussed the terms of the reporting system itself. All detailed requirements can be found in the table below (Table 6.1).

Table 6.1 The requirements for the dashboard, Source: Author

	The requirements for the dashboard
1	The dashboard was made to provide a general view and migrations of the diaspora structure in the Czech Republic.
2	The dashboard must be able to receive and show changes after the actualization of the data source.
3	Design should be simple, easily understood and user-friendly
4	The dashboard should dispose of charts and tables showing data about gender, residence, nationality, information, and reasons and period of migrations
5	The dashboard should contain filters that are referenced to every chart and table.

6.1.2 Data source

The main data source is the database table that is part of the SW solution for evidence of the Bosnian emigrants. Every category of the table is clarified in the previous chapter about the SW solution. This source was used solely for the dashboard solution in Excel because stored data is relatively simple and sufficient for building a dashboard. The version created in Power BI draws another source that contains the dimension tables. Those will be explained later in this chapter. Data imported in Power BI is from the Excel documents. All dimension tables are stored in one document, located on individual worksheets. Since the database already contains the final values, there was no need for data pre-processing.

6.1.3 Analysis of SW and HW equipment at the Embassy's office

The Embassy does not use any laptops, but every staff member has a computer with big monitors. All monitors have 21-inch displays and 1080p resolution, making them very suitable for the dashboard solution. This way, the author does not have to worry about the size of letters or charts used in the dashboard. Some of the employees had installed

Windows 7 as the operating system. Nevertheless, the author suggested reinstallation to Windows 10 and adding Microsoft Office 365 to ensure full functionality of both SW and BI solutions. Every side agreed on this action. Before testing and implementation, the author installed the Power BI Desktop version on every computer and added a date picker add-in to Excel.

6.1.4 Results of introductory study

After analysis of the steps of the introductory study, the author recognizes the possibility of implementing SW and BI solutions to the BiH Embassy in Prague. Requirements and needs can be fulfilled, and the realization is possible based on the data source and current SW and HW part of the Embassy.

6.2 Solution implemented in Power BI

This subchapter clarifies the process of implementing the solution in Power BI. It contains an identification of dimensions, data model, and design of the report. In this subchapter, the author describes charts used in Power BI, and places pictures from both programs for comparison.

6.2.1 Dimensions

For this part, the author presents all dimensions used for building a data model in Power BI software. All tables hold information about dimensions: id, dimension name, type, structure, attributes, and content. All dimensions came from the same Excel workbook and were, as such, imported.

Table 6.2 Dimension- citizenship, Source: Author

ID	DCitizenship
Dimension name	Permit
Type	Star
Attributes	Citizenship, id_permit, permit
Content	This dimension contains type of citizenship: BiH, CZ or double; and information about permits to which every type is connected to: visa, citizenship or no need for a visa.

Table 6.3 Dimension- Occupation, Source: Author

ID	DOccupation
Dimension name	Occupation
Type	Star
Attributes	Id_occupation, occupation, type_of_occupation
Content	This dimension contains information about fields of occupation that are grouped by the type of occupation. Three groups are defined: people oriented, sciences and governing.

Table 6.4 Dimension- Nationality, Source: Author

ID	DNationality
Dimension name	Declaration
Type	Star
Attributes	Id_nationality, nationality, declaration
Content	This dimension contains information about the types of nationality and, groups that they belong to. There are four types of nationality: Bosniak, Croatian, Serbian, and those belonging to the group of Declares. The group, "Does not declare" falls under "Other"

Table 6.5 Dimension- Education, Source: Author

ID	DEducation
Dimension name	Level of education
Type	Star
Attributes	Education, id_lvl_of_education, level of education
Content	This dimension contains information about the level of education. The types are: educated and uneducated.

6.2.2 Data model

The picture below shows the physical data model (Figure 6.1) (Annex 4). It was automatically created when importing the data sources. It contains one fact table named People and four dimension tables. The fact table is the table from the database, and every part was explained in the previous chapter. All relationships between fact and dimension table are 1..*. The only appropriate model type to use was a Star type, as shown in the picture below.

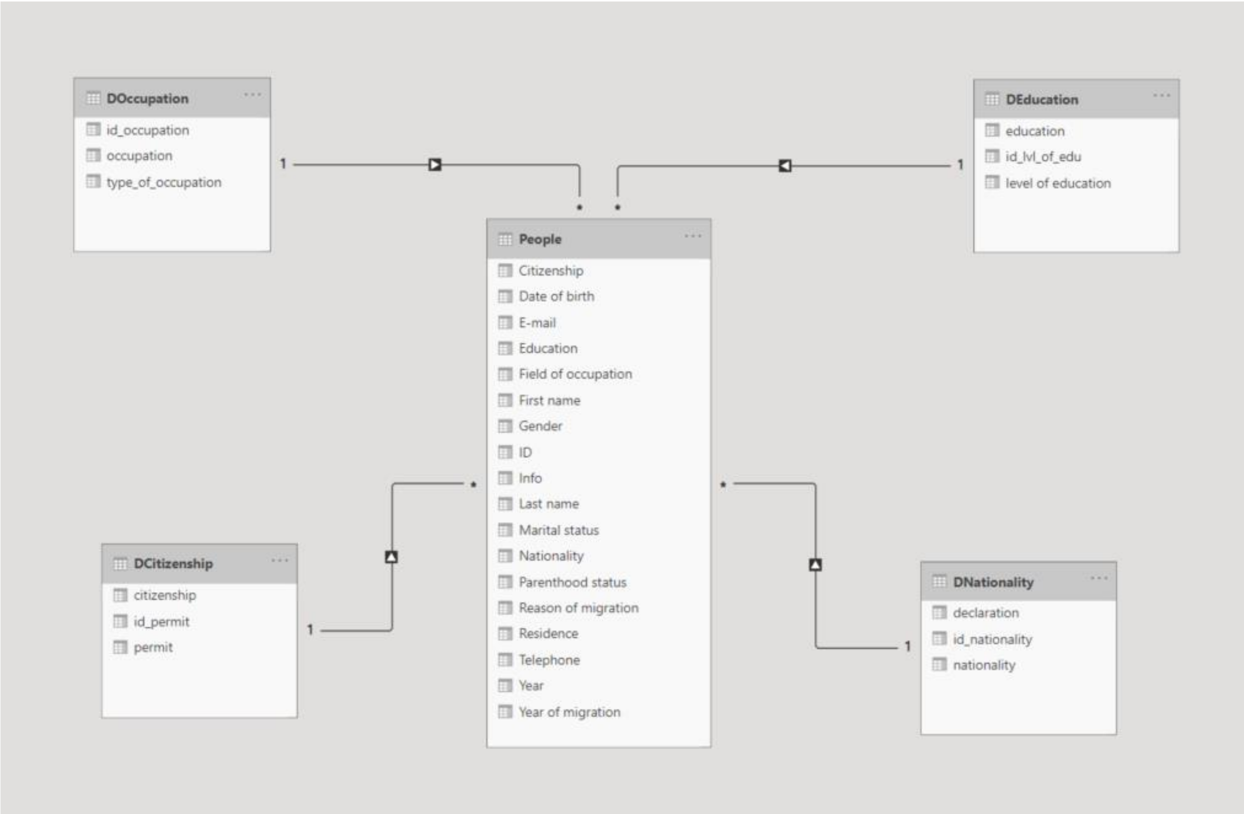


Figure 6.1 Physical data model in Power BI, Source: Author

6.2.3 Import and work with data

The author already explained data sources in the previous subchapter. Imported data is located in two workbooks. One workbook belongs to the SW solution explained in chapter four. The other workbook contains four dimension tables. Considering the author created the database table, it was already cleared up from all inequalities and ready for import. When importing this table, it was essential to choose just the table from the "Database" worksheet because the first row contains filters and buttons for data manipulation and would thus result in problems during the import and implementation. during the import and implementation. The author did not have to preprocess data because they were already sufficient as they are.

6.2.4 Creating reports

Power BI offers a great range of possible charts for making visuals. The author used various charts, tables, and slicers to meet the requirements stated by the Embassy. This solution is created of two dashboards. First gives a general view of the structure of the diaspora living in the Czech Republic. The second report provides information about migrations during the years. Every report was consulted with the Embassy's staff to ensure that they are up to expectations to ensure the fulfilment of the goals.

General structure view report

This report (Figure 6.2)(Annex 5) was based on the requirements of the employees of the Embassy. It is divided into three parts: left, central, and right, and each focuses on a different aspect.

The left part represents the general information about the emigrants, such as the number of people, gender, interest in info, and structure of people by regions in Czechia. The middle part is focused on statistics about the nationality of people and its declaration. The author decided to place slices in the middle part because of the aesthetics. Slicers used for this report are by residence, nationality, and gender, and they were decided upon by the end-users. The right part contains the information about the education of migrants.

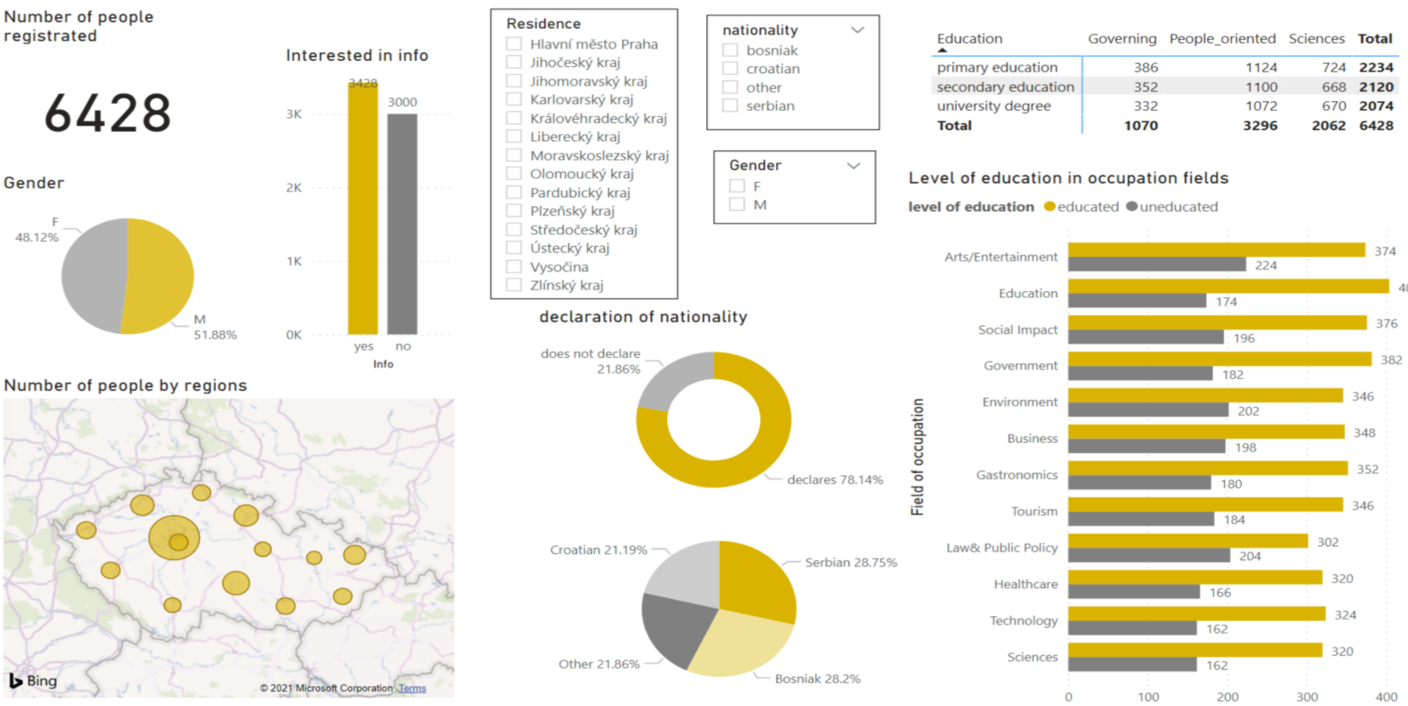


Figure 6.2 Structure view dashboard in Power BI, Source: Author

The author uses a Card (Figure 6.3) to represent the total number of registered people in the database.

Figure 6.3 Number of people registered- card, Source: Author

The partition by gender was represented using a pie chart (Figure 6.4)(Figure 6.5). A legend was not used because it contains only two categories that are labelled with category name and percentage. It shows that there are more male emigrants than female emigrants.

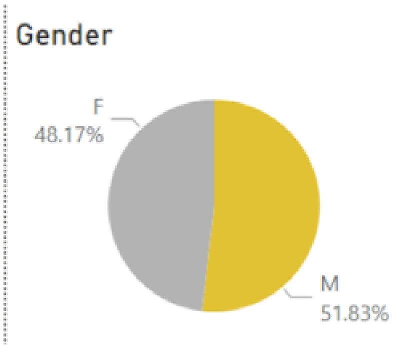


Figure 6.5 Gender - pie chart in Power BI, Source: Author

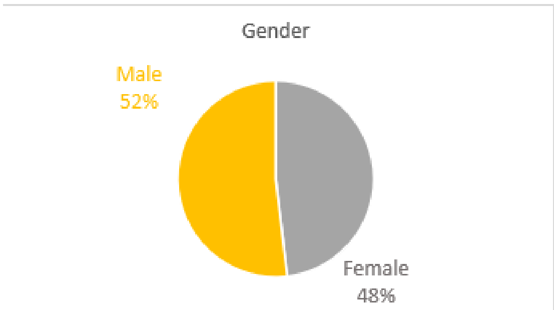


Figure 6.4 Gender - pie chart in Excel, Source: Author

Interest in info is represented by using a simple stacked column chart in Power BI (Figure 6.7) and bar chart in Excel (Figure 6.6). For better understanding, data labelling was turned on to show the number of people interested or disinterested in receiving information from the Embassy. It shows that more people expressed interest in receiving information than not.

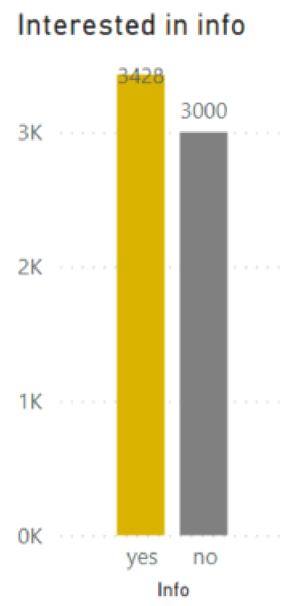


Figure 6.7 Info- stacked column chart in Power BI, Source: Author

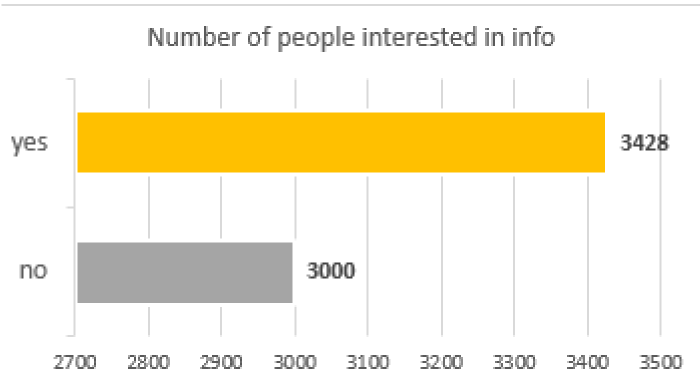


Figure 6.6 Info- bar chart in Excel, Source: Author

One of the requirements was to show the structure of emigrants by region, and for that purpose, the author used a map chart. The bigger circles in a Power BI (Figure 6.9) represent a larger percentage of people living in a specific region, and the darker coloring in Excel (Figure 6.8) represents the same thing. The region with the highest number of emigrants is Hlavní město Praha, and the second one is Vysočina. The region with the lowest number of emigrants is Olomoucký Kraj.

Figure 6.9 Residence- map chart in Power BI, Source: Author

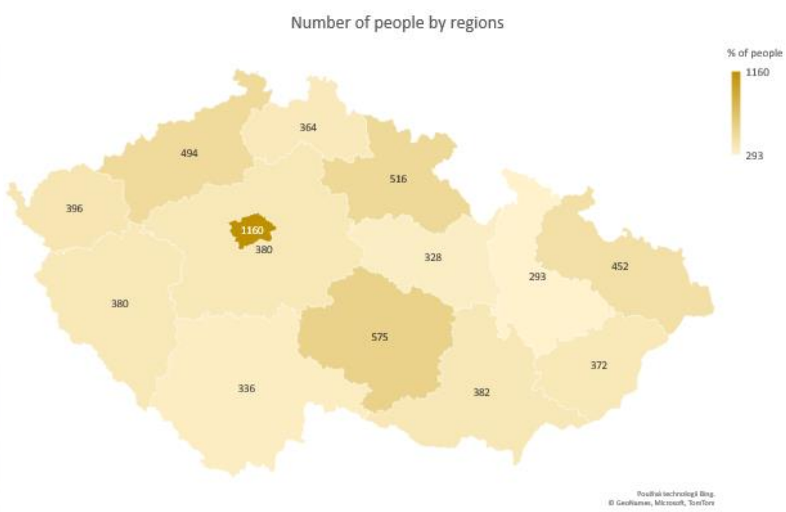


Figure 6.8 Residence- map chart in Excel, Source: Author

Declaration of nationality is presented using a donut chart (Figure 6.10). One can see that emigrants mainly declare their nationality and therefore belong to a specific group. This chart is not done in Excel, because declaration is on dimension created only for Power BI solution.

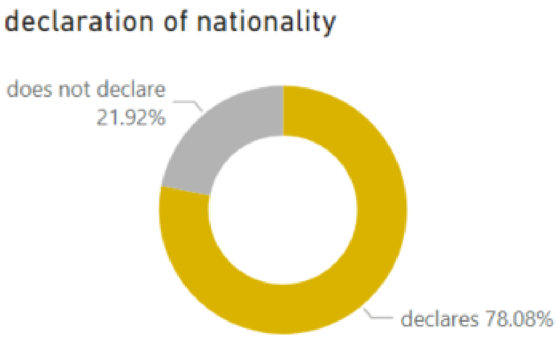


Figure 6.10 Declaration- donut chart in Power BI, Source: Author

The second pie chart (Figure 6.11) and (Figure 6.12) shows all four categories of nationality types. One can see that it is a more detailed version of the previous donut chart (Figure 6.10). It shows the percentage of people divided by nationality. It can be concluded that Serbs make up the largest group to migrate to the Czech Republic. Neither of the charts use a legend, but data is labelled with category names and percentages.

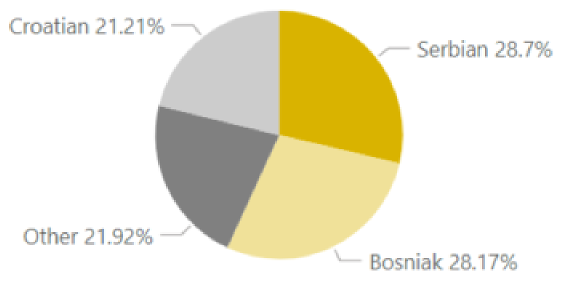


Figure 6.12 Nationality- pie chart in Power BI, Source: Author

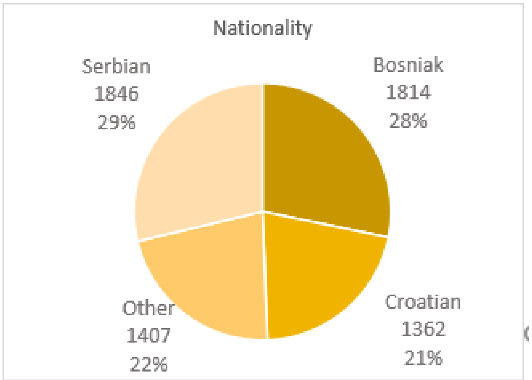


Figure 6.11 Nationality- pie chart in Excel, Source: Author

This report also consists of a table (Figure 6.13), showing the number of people, by their achieved level of education with the groups of occupation. It shows that most emigrants are doing people-oriented jobs, such as tourism, entertainment, or social impact. It also shows that more than half of people have a university degree.

Education	Governing	People_oriented	Sciences	**Total**
primary education	386	1124	724	**2234**
secondary education	352	1100	668	**2120**
university degree	332	1072	670	**2074**
Total	**1070**	**3296**	**2062**	**6428**

Figure 6.13 Education- table in Power BI, Source: Author

The last graph presented on a dashboard is used to show the level of education by occupation fields. It is a clustered bar chart (Figure 6.14), and it uses the legend to show two levels of education: educated and uneducated. This chart tells us that a significant majority is educated, but fields such as arts/entertainment and environment have a higher number of uneducated people.

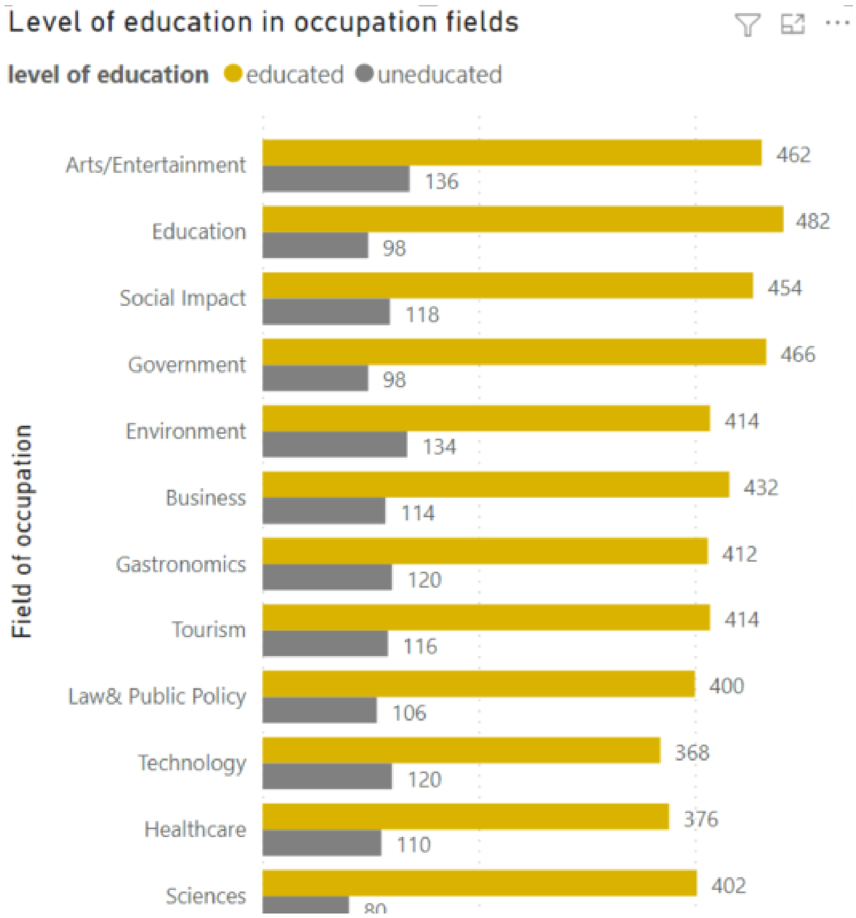


Figure 6.14 Education/occupation- stacked bar chart in Power Bi, Source: Author

The whole report can be filtered by residence, gender and nationality. For this, the author used slicers (Figure 6.15). The picture below shows the changes when some of the filters are turned on (Figure 6.16).

Figure 6.15 Slicers in Power BI, Source: Author

Figure 6.16 Filtered dashboard 1, Source: Author

Migration throughout the years report

The second report (Figure 6.17)(Annex 6) shows the migration throughout the periods. The Embassy wanted to know in which period most people migrated. The reason for this was a long war going on in Bosnia during the 90s, which made many people immigrate to other countries. They also wanted to know the reasons behind people's migrations to the Czech Republic. For example, if many young people are coming for studies, the Embassy could try to make more deals for scholarships of such students.

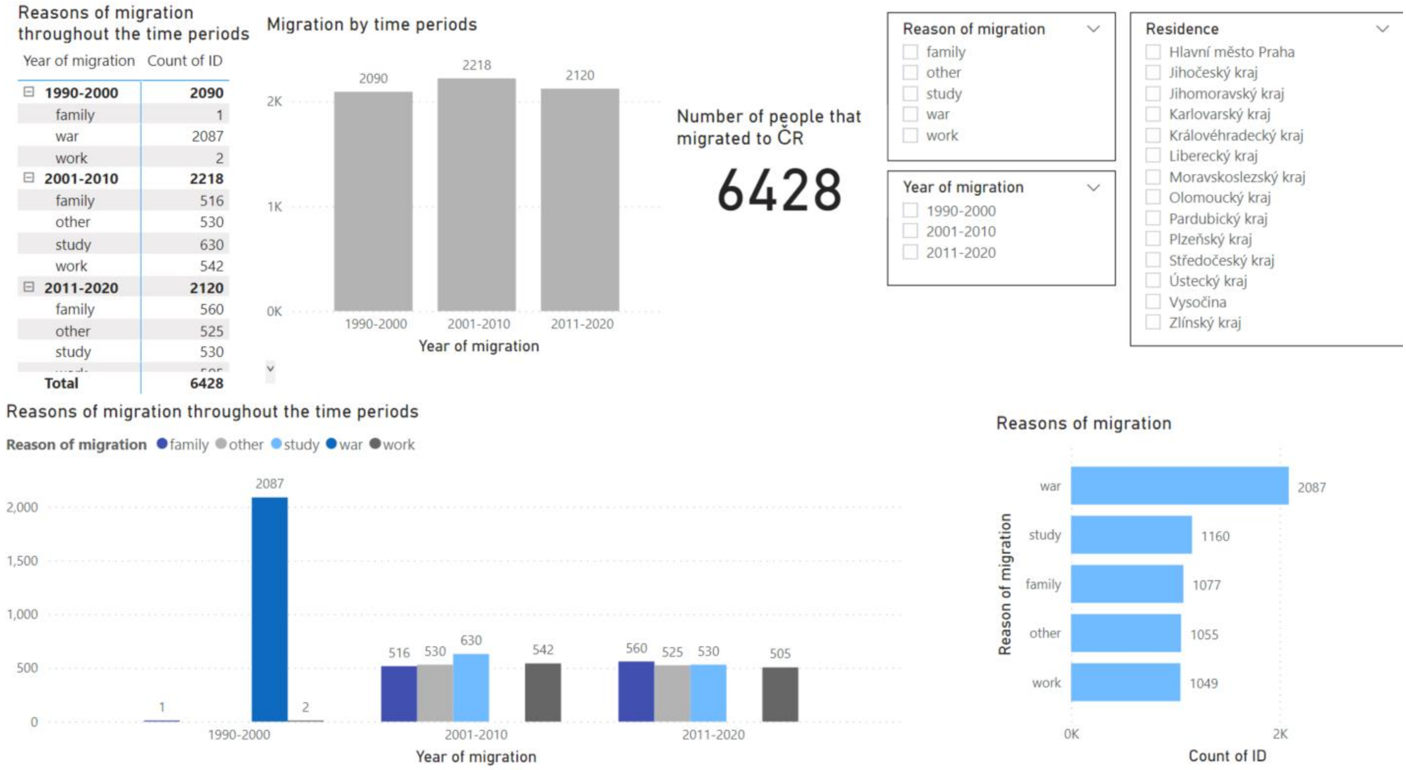


Figure 6.17 Migration throughout the years dashboard in Power BI, Source: Author

The left part shows the relation of time periods with the reasons behind the migrations. The middle part shows the total number of people who emigrated to Czechia from 1990 to 2020. This part also shows during which period most people moved to Czech Republic. The right part shows the reasons for emigration. The top right part consists of three slicers: by reason of migration, year of migration, and residence.

To represent the reasons of migration throughout the periods, the author decided to use both table and clustered column charts (See Figure 6.19 to Figure 6.21). This is because end-users stated that they prefer to see them both at the same report. These show that the greatest reason for the migration during 1990 was indeed war in Bosnia and Herzegovina. It can also be seen that fewer students moved to Czechia from 2011 to 2020 than during the period from 2001-2011.

Reasons of migration throughout the time periods

Year of migration	Count of ID
1990-2000	**2090**
family	1
war	2087
work	2
2001-2010	**2218**
family	516
other	530
study	630
work	542
2011-2020	**2120**
family	560
other	525
study	530
[illegible]	[illegible]
Total	**6428**

Figure 6.18 Reasons of migration- table in Power BI, Source: Author

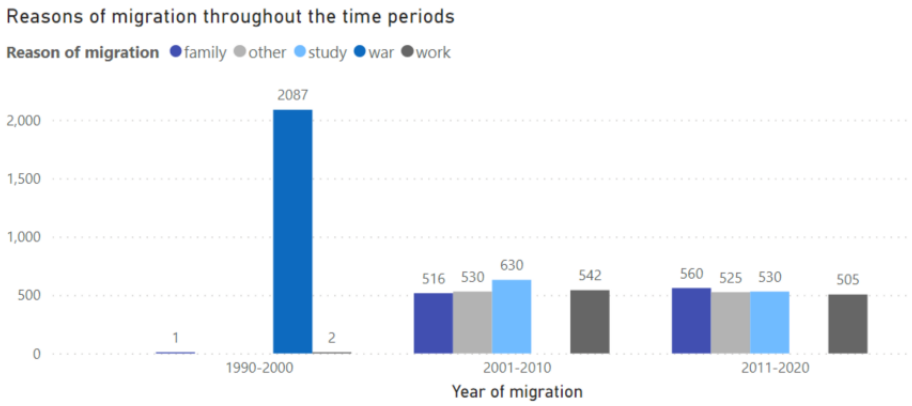


Figure 6.19 Reasons/ time periods- stacked column chart in Power BI, Source: Author

Year/reason of migration	Number of people
1990-2000	**2090**
family	1
war	2087
work	2
2001-2010	**2218**
family	516
other	530
study	630
work	542
2011-2020	**2120**
family	560
other	525
study	530
work	505
Total number of people	**6428**

Figure 6.21 Reasons of migration- table in Excel, Source: Author

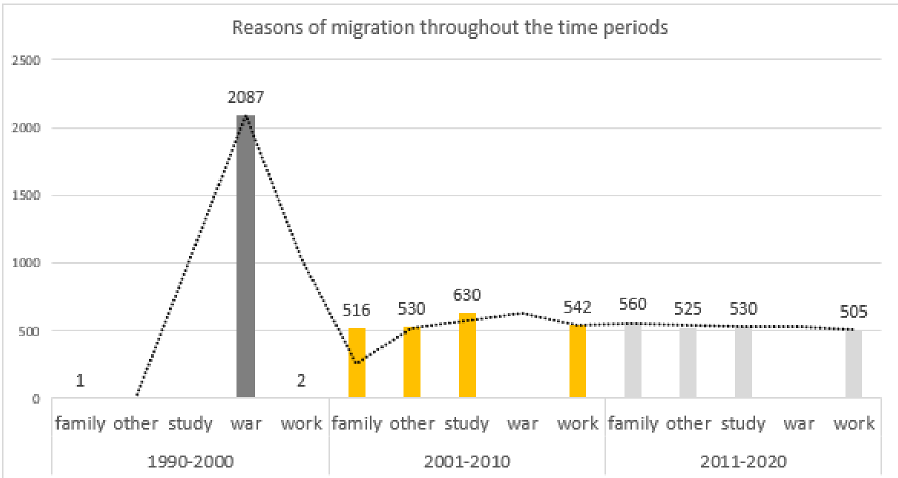


Figure 6.20 Reasons/ time periods- column chart in Excel, Source: Author

To visualize the migrations by time periods and determine which involved the most moving, the author used a stacked column chart. See (Figure 6.22) and (Figure 6.23). One can see that the period with the most migrations is the one from 2001 to 2010.

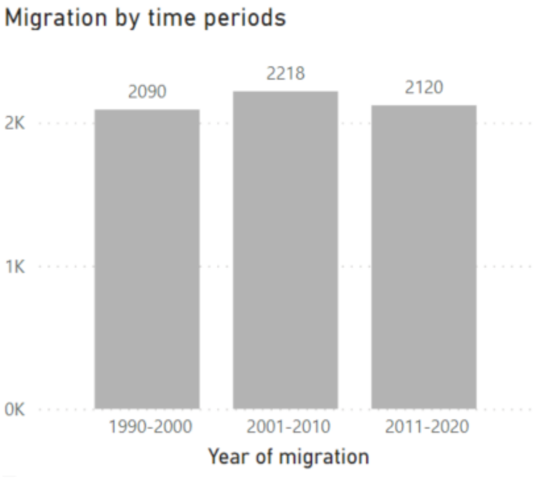


Figure 6.22 Year of migration: stacked column chart in Power BI, Source: Author

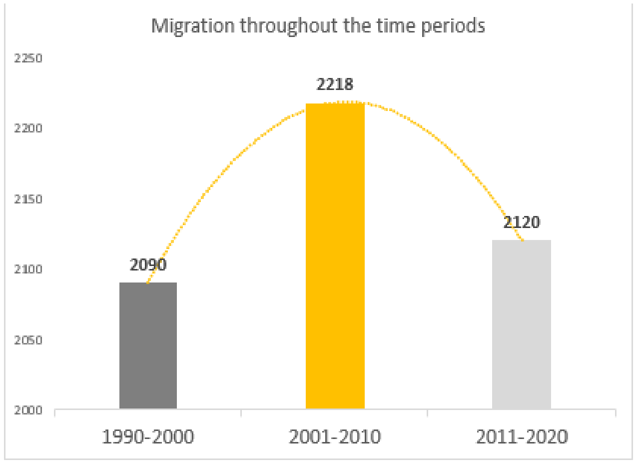


Figure 6.23 Year of migration: column chart in Excel, Source: Author

The fourth chart shows the reasons behind the migrations (Figure 6.24). For this purpose, the author used a bar chart and sorted categories by the most to least people per category. One can see that the greatest reason for migrations was the war, but many students are also coming to study in the Czech Republic. On the other side, the least people, only by a few percent, moved to Czechia because of the work.

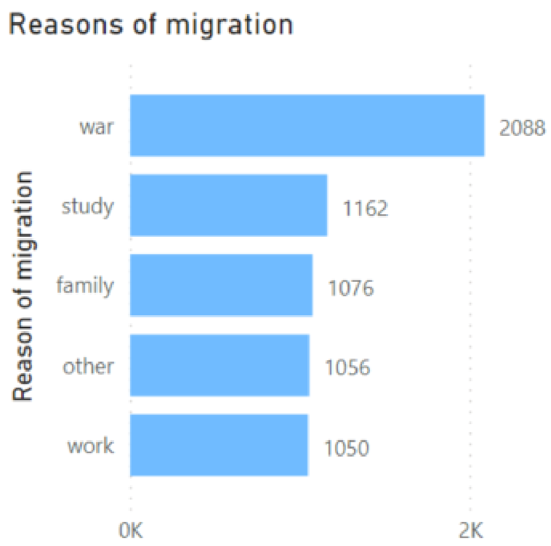


Figure 6.24 Reasons of migration- stacked bar chart in Power BI, Source: Author

Slicers were defined and required by the Embassy. The report can be filtered by reason and year of migration, as well as residence (Figure 6.25), so the staff could see which regions were mainly populated during the years.

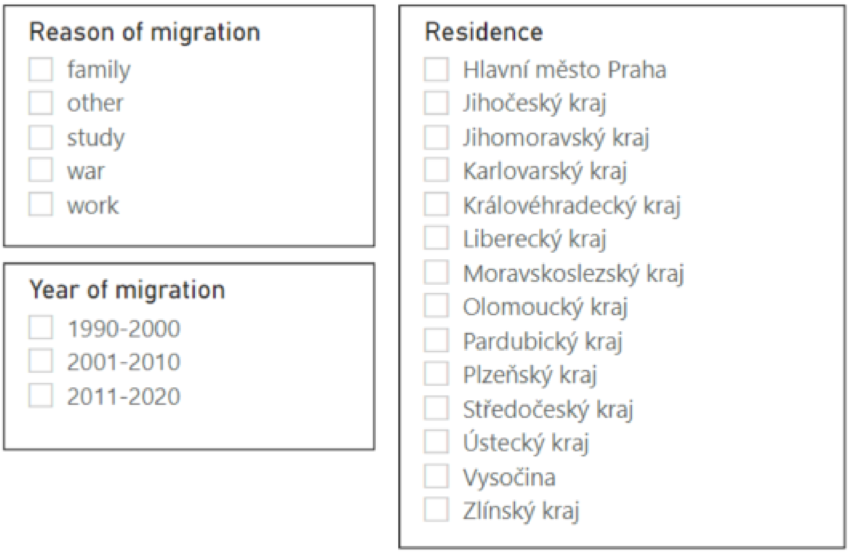


Figure 6.25 Slicers in Power BI, Source: Author

6.2.5 Actualization of data

To ensure that shown data reflects the database, it is essential to refresh and update dashboards. This can be done by clicking on the button "Refresh" both in Excel and Power BI. The users will be instructed and shown how to do this and find the right button. They will also be advised to execute this process every time they open a specific dashboard. Moreover, the author would explain why the dashboard should be actualized to ensure that the employees do not forget about it.

6.3 Solution built in Excel

For the solution created using Excel, the author did not have to create dimension tables and a data model. To create the dashboard (Figure 6.26) (Annex 7), the author used pivot tables. The most significant difference is that the Excel solution contains only one report, while the Power BI has two different reports. On the left, one can find slicers referenced and connected with every chart and table to enable the filtering throughout the whole dashboard. The top and right parts represent the general view of the diaspora structure. The bottom part is dedicated to migrations during the periods and reasons. The author uses the same charts as in the Power BI. Therefore, some are not individually clarified and only shown because they were already included in the subchapter of reports in Power BI.

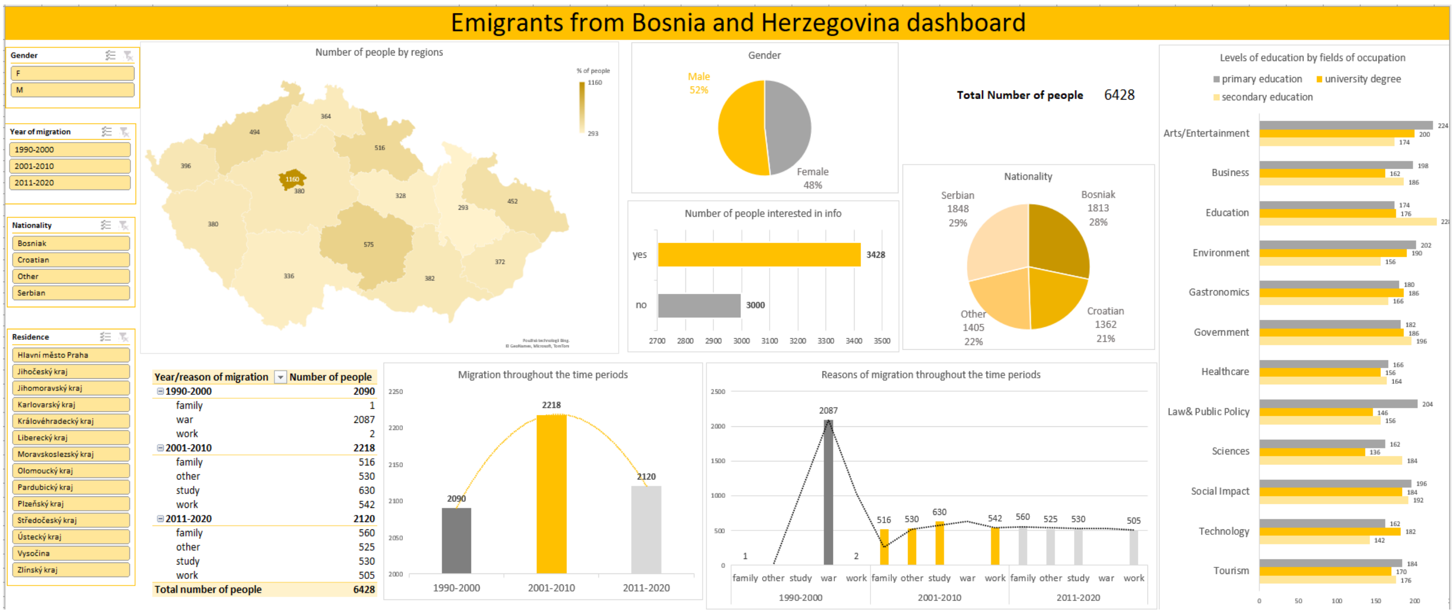


Figure 6.26 Emigrants from BiH dashboard in Excel, Source: Author

The chart that shows the level of education by the field of occupation is the stacked bar chart (Figure 6.27). The author decided to turn on the legend and data labelling for better understanding. One can see that the level of education remarkably differs from field to field. However, the highest percentage of emigrants with a university degree are in the environment and arts/entertainment fields.

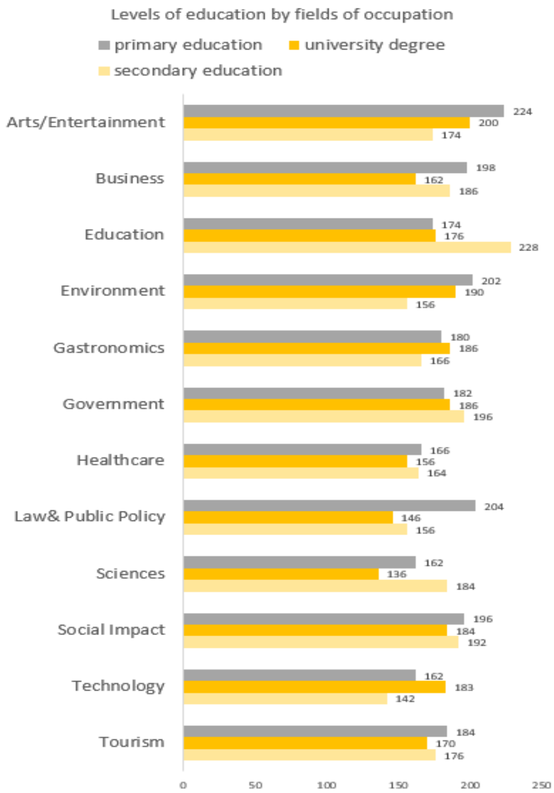


Figure 6.27 Level of education/occupation- bar chart in Excel, Source: Author

Slicers used for the dashboard solution in Excel are by gender, year of migration, nationality, and residence (Figure 6.28).

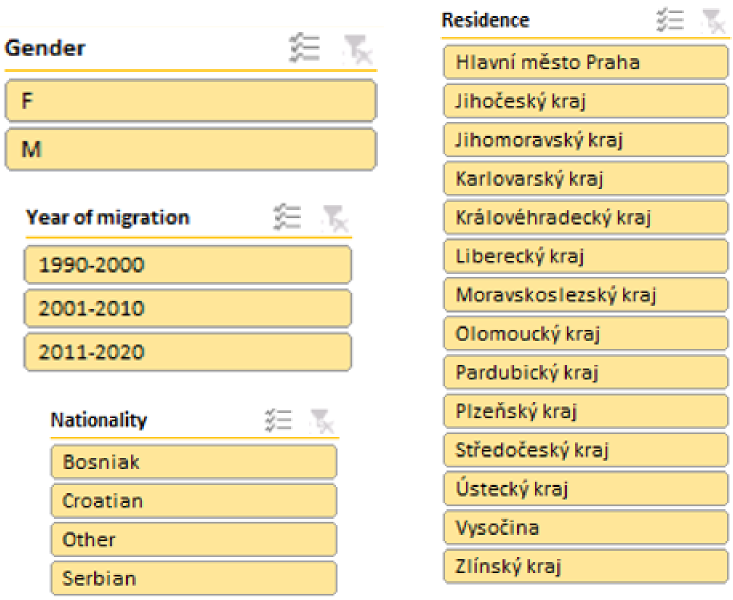


Figure 6.28 Slicers in Excel, Source: Author

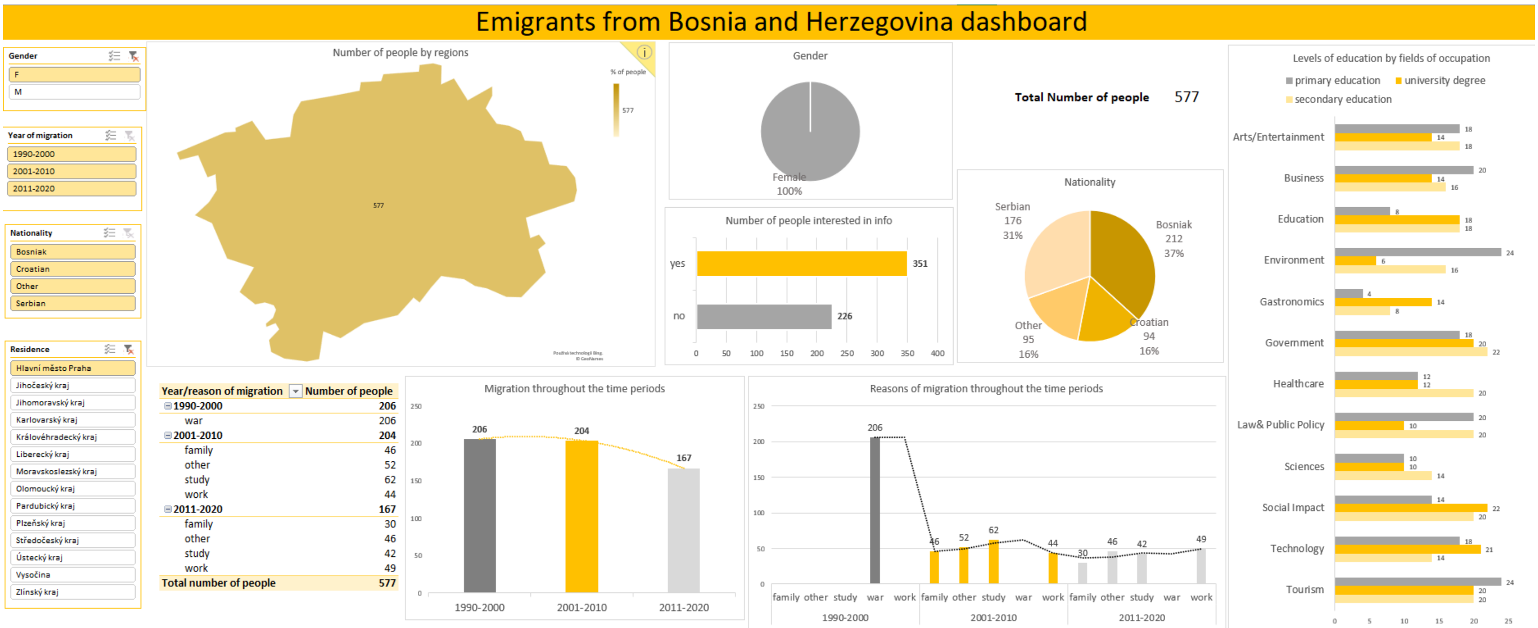


Figure 6.29 Filtered dashboard in Excel, Source: Author

The picture (Figure 6.29) above represents the changes that affect the dashboard when using slicers.

7 Testing the application and dashboard

Following the creation of the form and dashboard, it was necessary to test it by users. User testing will help gain insights from a different point of view, check the functionality and usability, and if they can be easily understood and used by the embassy's employees. Based on the outputs of testing, it is possible to adjust and edit the form and dashboard.

7.1 Aim of testing

The main goal of user testing is to verify if the form and dashboard components are effortlessly understandable to the end-users. It is also essential to check if it is user-friendly enough and if the users can use and work with it without prior experience and knowledge of the technology.

7.2 Test users

Five users participated in the testing process. All of them are current or former employees of the Bosnian Embassy in Prague. Former employees were used because most of them also worked/will work in different countries and could propose implementing the same system to the Bosnian embassies in different locations. The number of users is determined by the application and dashboard size, and all five have different profiles and knowledge. One could presume that the test results would be somewhat the same or similar and would not bring many differences if the number of test users was bigger.

Users that were chosen are the people who would work with the app and dashboard regularly. However, most of them do not have any prior knowledge of dashboards. Hence, they were chosen to prove if the dashboard can be handled and followed without problems.

7.3 User testing scenario for the form and dashboard

Testing was conducted on the work or private computers of test users with the OS Windows. Beforehand, it was advised to install and add the Date picker tool in Excel, so the form's calendar would be activated. The author suggested installation of Power BI as well. Each user received the application and dashboard documents, as well as questionnaire via email. For this part, the created document had to be filled in and sent back by the testers to gain better insights and recommendations. Testing was conducted via group call, where the author and the test users met. The goal of the call was to introduce the app and dashboard along with their functionalities, and it was an excellent opportunity for the author to get to answer direct questions about the project. From this, the author was able to gain insight on what was not understood by users and what should be changed. The goal was to ensure that all testers understand all components, categories, and graphs that were part of the system. After introducing every part and possibilities, each user got a few questions to answer about the meaning of some parts of the app or dashboard. This was a way for the author to check if everything was understood before using testing data, but also, it is essential that users can follow the system. Afterward, they were given time to try them out using recommended testing data, or data on their own, on their computers, and fill in the questionnaire's specific parts. It was tested for whether the end-user can pick up on the rules of data entries and functionality of the filters on the form and database table. It was also tested if they knew what the graphs meant in the dashboard. They did not have to send reviews back immediately but had few days to test the app and dashboard again and finish the questionnaire. Both questions during the call and review will give even better insight into problems with the project. Once filled in forms are sent in, the author can start analysing the answers and gain the testing results.

7.4 Testing results

7.4.1 Testing of a SW solution

Testing results came out very positive. All attendants were able to understand every category of data and various filters. They understood the purpose of every button, and they used them very intuitively. For example, the process of updating a record did not cause any problems to the testers, and each person knew how to delete a row in an Excel table. This was important information for the author since the processes of Updating and Deleting records were based on the basic knowledge of Excel by employees.

During the video call, attendants were asked to confirm if every data validation was working correctly, and each verified as yes. They were also asked to verify if every button and filter works correctly, and this question was also confirmed as yes. However, the biggest problem and misunderstanding for the employees were the drop-down filters. They did not expect them to filter data immediately and mainly tried to find a button to confirm the filter. However, after testing it, they said that it is a big plus that they do not have to confirm filters when selecting from the drop-down lists. Another cause of confusion was the locked worksheet because some tried to click on the data in the table, but after the author explained the reason behind the locked worksheet, they accepted the arguments.

7.4.2 Testing of a dashboard solutions

During the testing of a dashboard, attendants were able to read and understand the purpose of every chart used on a dashboard. When asked to actualize data, they instinctively knew where to click in Excel and Power BI. They prefer having two dashboards with different goals; thus, they were more optimistic about the option in the Power BI. They were asked to verify if all slicers work and filter through the dashboard, and this was confirmed as yes. They also stated that the Power BI solution is better because it gives even more information, for example: how nationality is grouped by declaration. Two testers had to take time to understand the chart "Reasons of migration throughout the time periods." It was mainly because they needed time to understand the legend and colors. They particularly liked the "General view of people" dashboard and said that it gives a good insight into the structure of the Bosnian diaspora in the Czech Republic. They liked that the table was being used on the "Migration throughout the years" report because it shows the information differently than just the charts. At the end of the testing, it was agreed that the Power BI solution is the one that should be used. They also stated that it would be a good idea to have minor training for working in Power BI, so they could be able to make and edit reports by themselves in the future if the solution gets implemented.

7.4.3 Questionnaire answers

Apart from testing during a video call, all employees were asked to fill in the questionnaire about the functionality of the SW and BI solution and give some of their opinions. Questions about the functionality of buttons, filters, and basic functionalities were 100% answered as yes. Everyone also agreed that it is very user-friendly and intuitive and stated that it is safe enough to be used. They liked the option of having a password to make very hidden worksheets visible and also liked that they were locked so no one could accidentally make mistakes inside them. The most mixed answers were on the question: "Is there enough information presented?" (Figure 7.1). 40% thought that the already shown information in the dashboard was enough, while the other 40% wished to see even more categories, like including citizenship and marital status.

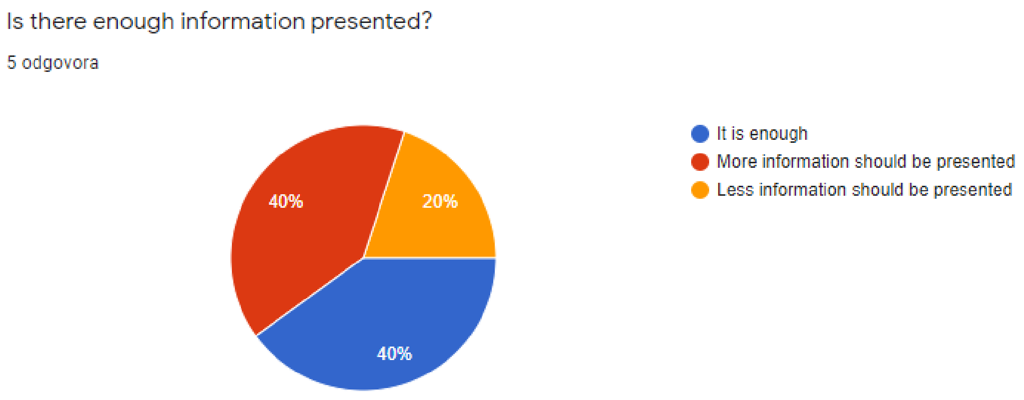


Figure 7.1 Questionnaire question - presented information, Source: Author

Nevertheless, this was just a proposal for the future, if there is a need for including such categories. Only 20% thought that there was already too much information represented. The same 20% voted for the solution in Excel (Figure 7.2), while other 80% stated that the Power BI solution is much better and more valuable. Therefore, if the Embassy decides to implement the application into their system, the used software for the graphic display would be Power BI. They were also asked to say their opinion in a few sentences (Figure 7.3). Employees wished to be anonymous. Therefore, the author does not include names or collects email addresses.

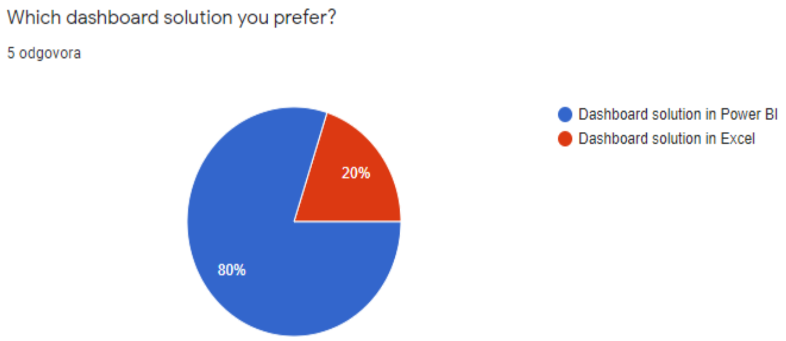


Figure 7.2 Questionnaire question- preferred dashboard solution, Source: Author

Please give your honest opinion of the SW solution and dashboard. Use 2 to 3 sentences and write in English.

5 odgovora

I think it is good

It is very good and useful, especially for having in one place all necessary information about diaspora of Bosnia and Herzegovina in Czech Republic. It can be useful for keeping good relations of diaspora with home country, especially for youngsters. Also could be useful for developing good relations in fields of economy, culture and politics.

The application can be upgraded and used for economic diplomacy (BiH Diaspora Investment Survey in BiH) for elections (sending information and voting materials) for consular affairs (consular issues)

its very helpful to have such data base. It will be very helpful for all Bosnian diplomatic mission around the world

This application is very useful and fulfilles my expectations. Dashboard shows everything I wanted to see.

Figure 7.3 Questionnaire question - opinion about the solutions, Source: Author

7.5 Conclusion of testing

During the video meeting, the author did not get many testers' questions because mostly everything was easily understandable. Most questions asked were about the data categories and how some of the filters work. They also wanted to know if the application and dashboard could be further edited and developed, like adding more filters, categories, and more dashboards for different purposes. However, this question was meant for the distant future, if the Embassy implements these solutions to their systems. Some of the testers had other ideas for which these solutions can be used, as shown in the picture (Figure 7.3).

In conclusion, the author was able to meet the requirements and goals stated by the employees. Every part was easily understandable, and the functionality was confirmed. None of the attendants had any significant issues during the testing, which ensures the author that the solution could be implemented the way it already is. Also, testers decided that the dashboard in Power BI is a better solution, therefore it could be implemented in the future.

Conclusions

This thesis's primary purpose was to create and implement an SW solution for reporting and registration of Bosnian emigrants in the Czech Republic for the Embassy of Bosnia and Herzegovina in Prague. The author recognizes that having such a system would be a beneficial and great way to help the Embassy move towards digitalization and make evidence of the records and reporting more effective. It will be helpful for the Embassy employees, whose job is to keep and manage records. Furthermore, it will be useful to get insights and follow the structure of the Bosnian diaspora using information represented on dashboards.

The first part of this thesis is devoted to the theoretical part. In this part, the author is introducing the fundamentals and concepts of data visualization and dashboards. The author also describes the principles of building a dashboard while covering the correct ways of using visuals. One chapter is dedicated solely to the Embassy of Bosnia and Herzegovina in Prague to help the reader gain insight into their missions and problematics.

The second segment is dedicated to the practical part of this thesis. The author uses the principles defined in the theoretical claim for the correct creation of the SW solution and its documentation. Firstly, the author analyses the needs of the Embassy for a better understanding and setting up of the right goals and requirements for the solution. It consists of two parts: SW application and dashboard. To create the application, the author used her prior knowledge gained at the university and work. Dashboard visuals were created by the principles explained in the first part of the thesis.

The last part is dedicated to the testing results of the SW solution. In this section, the author uses Embassy employees to verify the functionality and get better insight into their opinions. This part helps the author decide which version of the dashboard solution to choose for implementation.

This thesis provides an analysis of the needs of the Bosnian Embassy in Prague and contains complete documentation of the implementation of the SW and dashboard solution. The confirmation of delivery of the solution is in the near future, and it is up to the author and the Embassy to continue further development and implementation.

List of references

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Annexes

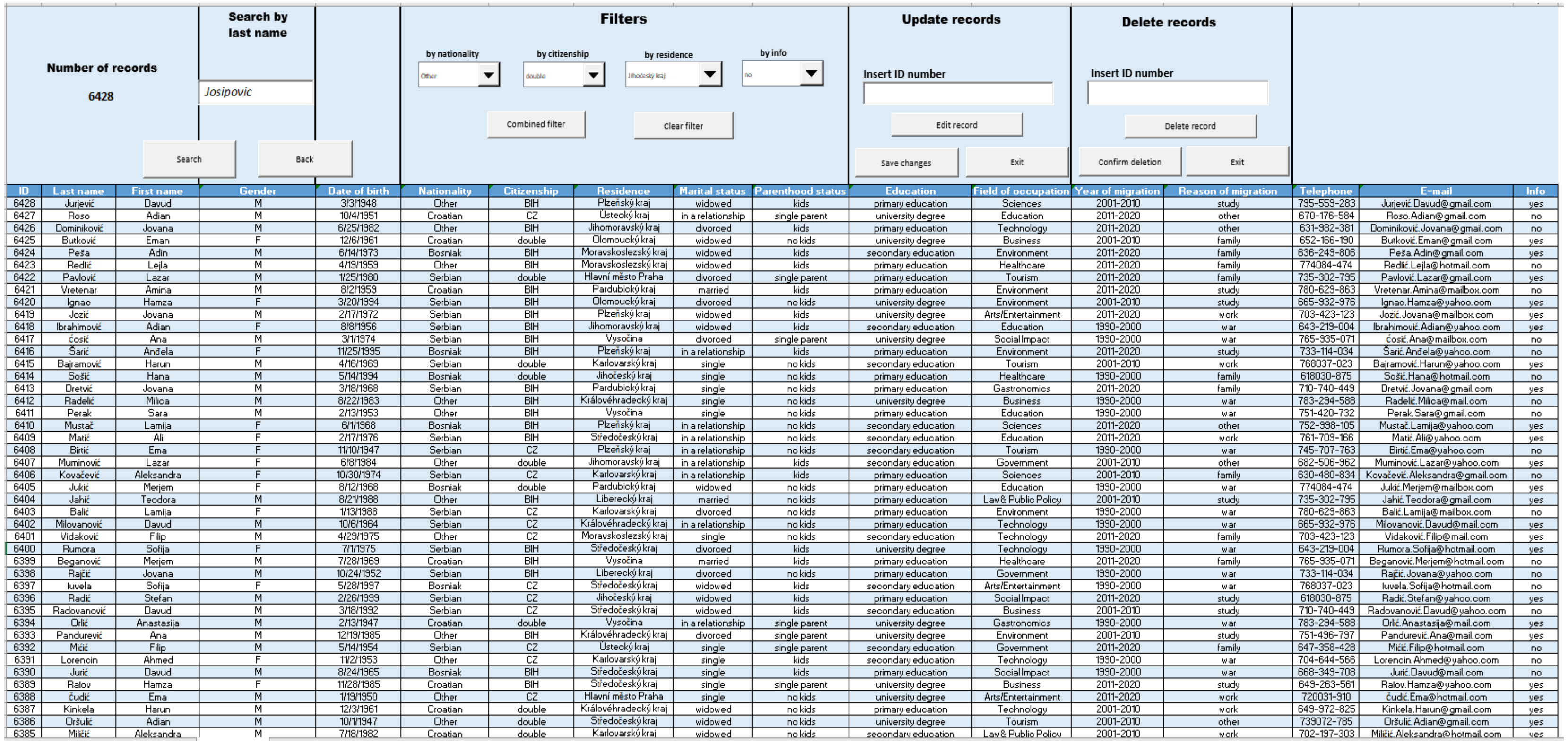
Annex A: Visualization of “Password” worksheet in Excel

Annex 1, Source: Author

Annex B: Visualization of Data entry from in Excel

Annex 2 Source: Author

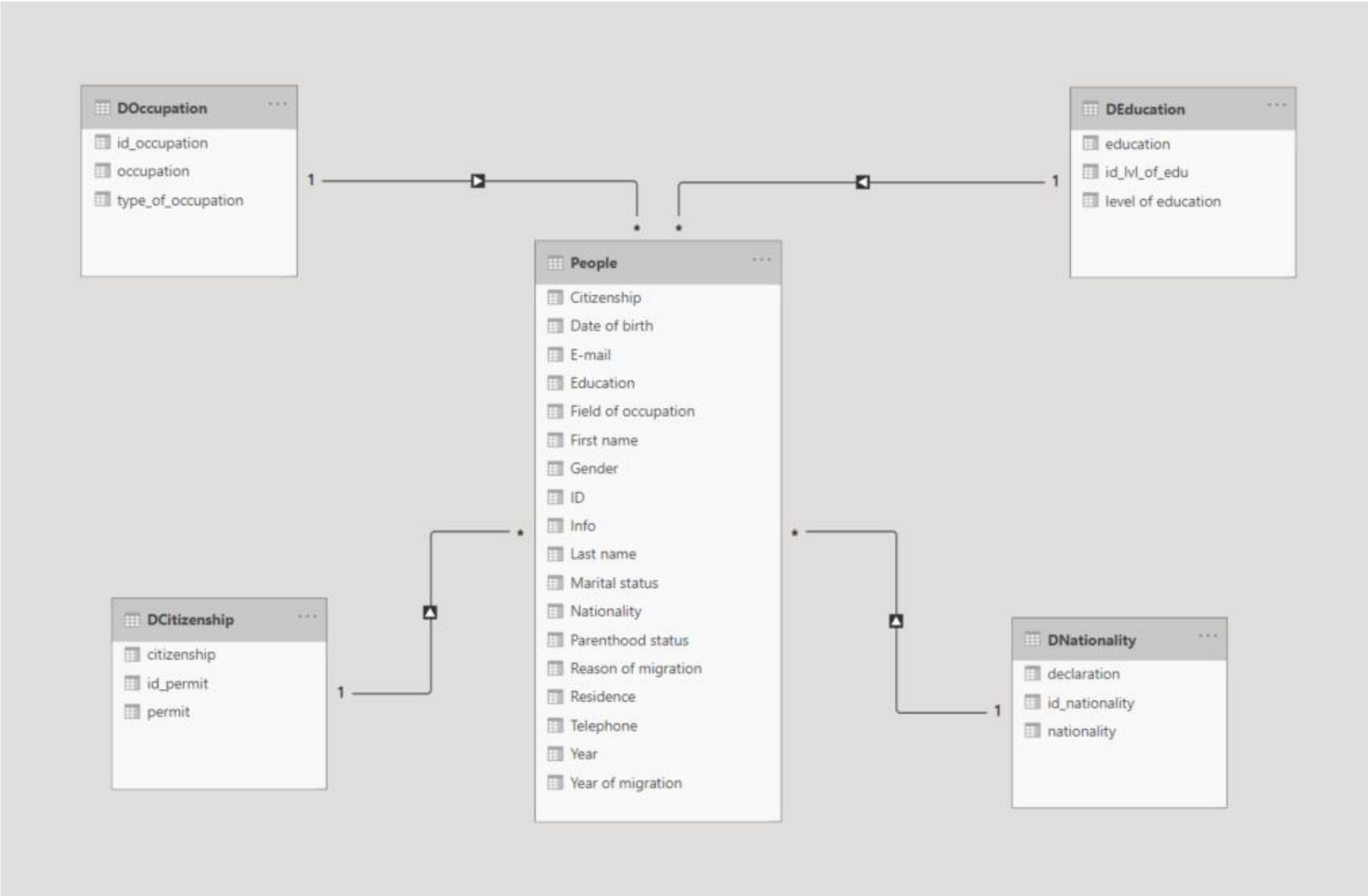
Annex C: Visualization of Database table in Excel



ID	Last name	First name	Gender	Date of birth	Nationality	Citizenship	Residence	Marital status	Parenthood status	Education	Field of occupation	Year of migration	Reason of migration	Telephone	E-mail	Info
6428	Jurjević	Davud	M	3/3/1948	Other	BIH	Plzeňský kraj	widowed	kids	primary education	Sciences	2001-2010	study	795-559-283	Jurjević.Davud@gmail.com	yes
6427	Roso	Adian	M	10/4/1951	Croatian	CZ	Ústecký kraj	in a relationship	single parent	university degree	Education	2011-2020	other	670-176-584	Roso.Adian@gmail.com	no
6426	Dominiković	Jovana	M	6/25/1982	Other	BIH	Jihomoravský kraj	divorced	kids	primary education	Technology	2011-2020	other	631-982-381	Dominiković.Jovana@gmail.com	no
6425	Butković	Eman	F	12/6/1961	Croatian	double	Olomoucký kraj	widowed	no kids	university degree	Business	2001-2010	family	652-166-190	Butković.Eman@gmail.com	yes
6424	Peša	Adin	M	6/14/1973	Bosniak	BIH	Moravskoslezský kraj	widowed	kids	secondary education	Environment	2011-2020	family	636-249-806	Peša.Adin@gmail.com	yes
6423	Redlić	Lejla	M	4/19/1959	Other	BIH	Moravskoslezský kraj	widowed	kids	primary education	Healthcare	2011-2020	family	774084-474	Redlić.Lejla@hotmail.com	no
6422	Pavlović	Lazar	M	1/25/1980	Serbian	double	Hlavní město Praha	divorced	single parent	primary education	Tourism	2011-2020	family	735-302-795	Pavlović.Lazar@gmail.com	yes
6421	Vretenar	Amina	M	8/2/1959	Croatian	BIH	Pardubický kraj	married	kids	primary education	Environment	2011-2020	study	780-629-863	Vretenar.Amina@mailbox.com	no
6420	Ignac	Hamza	F	3/20/1994	Serbian	BIH	Olomoucký kraj	divorced	no kids	university degree	Environment	2001-2010	study	665-932-976	Ignac.Hamza@yahoo.com	yes
6419	Jozić	Jovana	M	2/17/1972	Serbian	BIH	Plzeňský kraj	widowed	kids	university degree	Arts/Entertainment	2011-2020	work	703-423-123	Jozić.Jovana@mailbox.com	yes
6418	Ibrahimović	Adian	F	8/8/1956	Serbian	BIH	Jihomoravský kraj	widowed	kids	secondary education	Education	1990-2000	war	643-219-004	Ibrahimović.Adian@yahoo.com	yes
6417	ćosić	Ana	M	3/1/1974	Serbian	BIH	Vysočina	divorced	single parent	university degree	Social Impact	1990-2000	war	765-935-071	ćosić.Ana@mailbox.com	no
6416	Šarić	Anđela	F	11/25/1995	Bosniak	BIH	Plzeňský kraj	in a relationship	kids	primary education	Environment	2011-2020	study	733-114-034	Šarić.Anđela@yahoo.com	no
6415	Bajramović	Harun	M	4/16/1969	Serbian	double	Karlovarský kraj	single	no kids	secondary education	Tourism	2001-2010	work	768037-023	Bajramović.Harun@yahoo.com	yes
6414	Sošić	Hana	M	5/14/1994	Bosniak	double	Jihočeský kraj	single	no kids	primary education	Healthcare	1990-2000	family	618030-875	Sošić.Hana@hotmail.com	no
6413	Dretvić	Jovana	M	3/18/1968	Serbian	BIH	Pardubický kraj	single	no kids	primary education	Gastronomics	2011-2020	family	710-740-449	Dretvić.Jovana@gmail.com	yes
6412	Radelić	Milica	M	8/22/1983	Other	BIH	Královéhradecký kraj	single	no kids	university degree	Business	1990-2000	war	783-294-588	Radelić.Milica@mail.com	no
6411	Perak	Sara	M	2/13/1953	Other	BIH	Vysočina	single	no kids	primary education	Education	1990-2000	war	751-420-732	Perak.Sara@gmail.com	no
6410	Mustač	Lamija	F	6/1/1968	Bosniak	BIH	Plzeňský kraj	in a relationship	no kids	secondary education	Sciences	2011-2020	other	752-998-105	Mustač.Lamija@yahoo.com	yes
6409	Matić	Ali	F	2/17/1976	Serbian	BIH	Středočeský kraj	in a relationship	no kids	secondary education	Education	2011-2020	work	761-709-166	Matić.Ali@yahoo.com	yes
6408	Birtić	Ema	F	11/10/1947	Serbian	CZ	Plzeňský kraj	in a relationship	no kids	secondary education	Tourism	1990-2000	war	745-707-763	Birtić.Ema@yahoo.com	no
6407	Muminović	Lazar	F	6/8/1984	Other	double	Jihomoravský kraj	in a relationship	kids	secondary education	Government	2001-2010	other	682-506-962	Muminović.Lazar@yahoo.com	yes
6406	Kovačević	Aleksandra	F	10/30/1974	Serbian	CZ	Karlovarský kraj	in a relationship	kids	primary education	Sciences	2001-2010	family	630-480-834	Kovačević.Aleksandra@gmail.com	no
6405	Jukić	Merjem	F	8/12/1968	Bosniak	double	Pardubický kraj	widowed	no kids	primary education	Education	1990-2000	war	774084-474	Jukić.Merjem@mailbox.com	yes
6404	Jahić	Teodora	M	8/21/1988	Other	BIH	Liberecký kraj	married	no kids	primary education	Law& Public Policy	2001-2010	study	735-302-795	Jahić.Teodora@gmail.com	yes
6403	Balić	Lamija	F	1/13/1988	Serbian	CZ	Karlovarský kraj	divorced	no kids	primary education	Environment	1990-2000	war	780-629-863	Balić.Lamija@mailbox.com	no
6402	Milovanović	Davud	M	10/6/1964	Serbian	CZ	Královéhradecký kraj	in a relationship	no kids	primary education	Technology	1990-2000	war	665-932-976	Milovanović.Davud@mail.com	yes
6401	Vidaković	Filip	M	4/29/1975	Other	CZ	Moravskoslezský kraj	single	no kids	secondary education	Technology	2011-2020	family	703-423-123	Vidaković.Filip@mail.com	yes
6400	Rumora	Sofija	F	7/1/1975	Serbian	BIH	Středočeský kraj	divorced	kids	university degree	Technology	1990-2000	war	643-219-004	Rumora.Sofija@hotmail.com	yes
6399	Beganović	Merjem	M	7/28/1969	Croatian	BIH	Vysočina	married	kids	primary education	Healthcare	2011-2020	family	765-935-071	Beganović.Merjem@hotmail.com	no
6398	Rajčić	Jovana	M	10/24/1952	Serbian	BIH	Liberecký kraj	divorced	no kids	primary education	Government	1990-2000	war	733-114-034	Rajčić.Jovana@yahoo.com	no
6397	Iuvela	Sofija	F	5/28/1997	Bosniak	CZ	Středočeský kraj	widowed	kids	secondary education	Arts/Entertainment	1990-2000	war	768037-023	Iuvela.Sofija@hotmail.com	no
6396	Radić	Stefan	M	2/26/1999	Serbian	CZ	Jihočeský kraj	widowed	kids	primary education	Social Impact	2011-2020	study	618030-875	Radić.Stefan@yahoo.com	yes
6395	Radovanović	Davud	M	3/18/1992	Serbian	CZ	Středočeský kraj	widowed	kids	secondary education	Business	2001-2010	study	710-740-449	Radovanović.Davud@yahoo.com	no
6394	Orlić	Anastasija	M	2/13/1947	Croatian	double	Vysočina	in a relationship	single parent	university degree	Gastronomics	1990-2000	war	783-294-588	Orlić.Anastasija@mail.com	yes
6393	Pandurević	Ana	M	12/19/1985	Other	BIH	Královéhradecký kraj	divorced	single parent	university degree	Environment	2001-2010	study	751-496-797	Pandurević.Ana@mail.com	yes
6392	Mićić	Filip	M	5/14/1954	Serbian	CZ	Ústecký kraj	single	single parent	secondary education	Government	2011-2020	family	647-358-428	Mićić.Filip@hotmail.com	no
6391	Lorencin	Ahmed	F	11/2/1953	Other	CZ	Karlovarský kraj	single	kids	secondary education	Technology	1990-2000	war	704-644-566	Lorencin.Ahmed@yahoo.com	no
6390	Jurić	Davud	M	8/24/1965	Bosniak	BIH	Středočeský kraj	single	kids	primary education	Social Impact	1990-2000	war	668-349-708	Jurić.Davud@mail.com	no
6389	Ralov	Hamza	F	11/28/1985	Croatian	BIH	Středočeský kraj	single	single parent	university degree	Business	2011-2020	study	649-263-561	Ralov.Hamza@yahoo.com	yes
6388	čudić	Ema	M	1/19/1950	Other	CZ	Hlavní město Praha	single	no kids	university degree	Arts/Entertainment	2011-2020	work	720031-910	čudić.Ema@hotmail.com	yes
6387	Kinkela	Harun	M	12/3/1961	Croatian	double	Královéhradecký kraj	widowed	no kids	primary education	Technology	2001-2010	work	649-972-825	Kinkela.Harun@gmail.com	yes
6386	Oršulić	Adian	M	10/1/1947	Other	double	Středočeský kraj	widowed	no kids	university degree	Tourism	2001-2010	other	739072-785	Oršulić.Adian@gmail.com	yes
6385	Miličić	Aleksandra	M	7/18/1982	Croatian	double	Karlovarský kraj	widowed	no kids	secondary education	Law& Public Policy	2001-2010	work	702-197-303	Miličić.Aleksandra@hotmail.com	yes

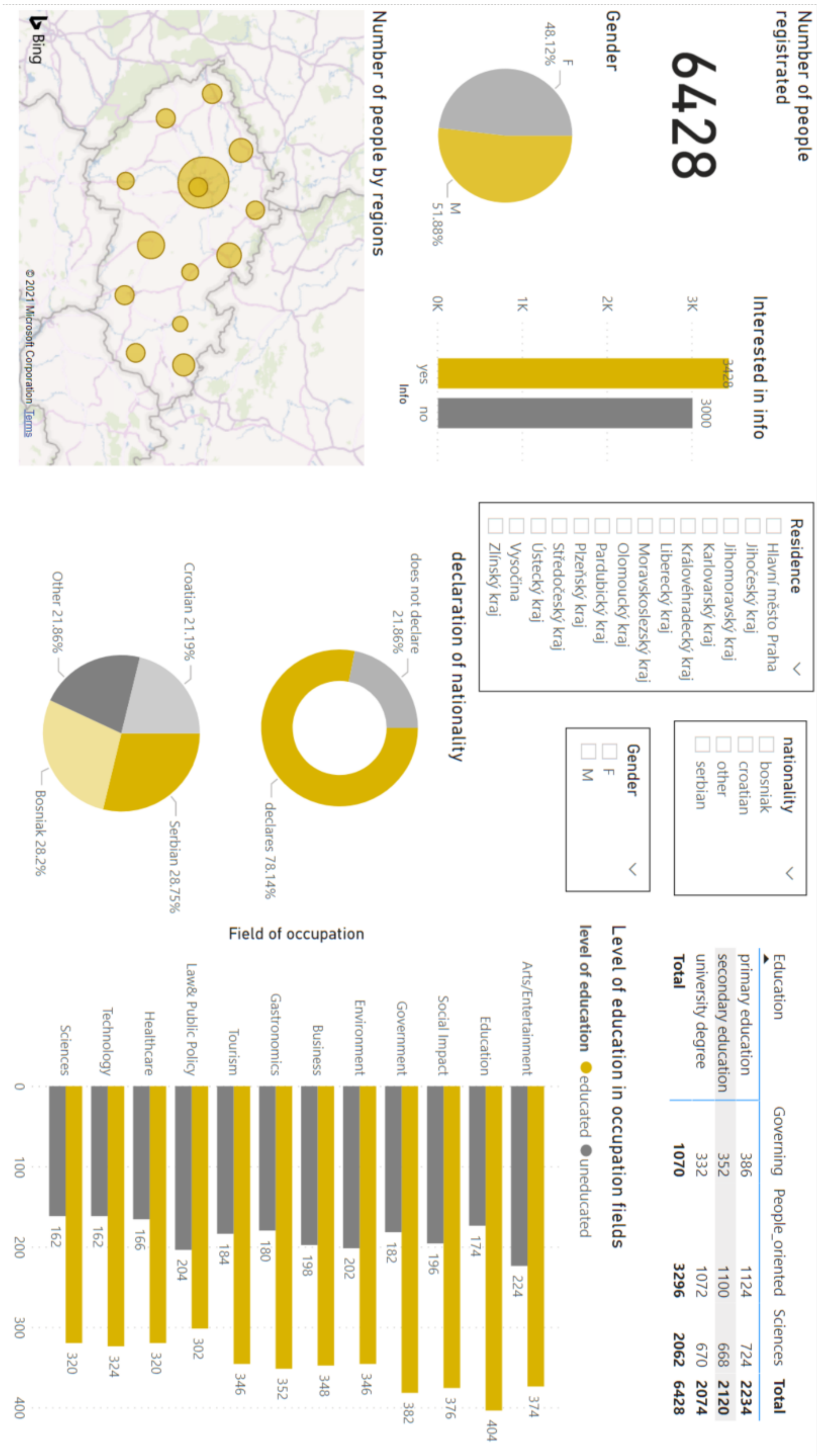
Annex 3 Source: Author

Annex D: Data model



Annex 4 Source: Author

Annex E: Dashboard “General view of structure” in PBI

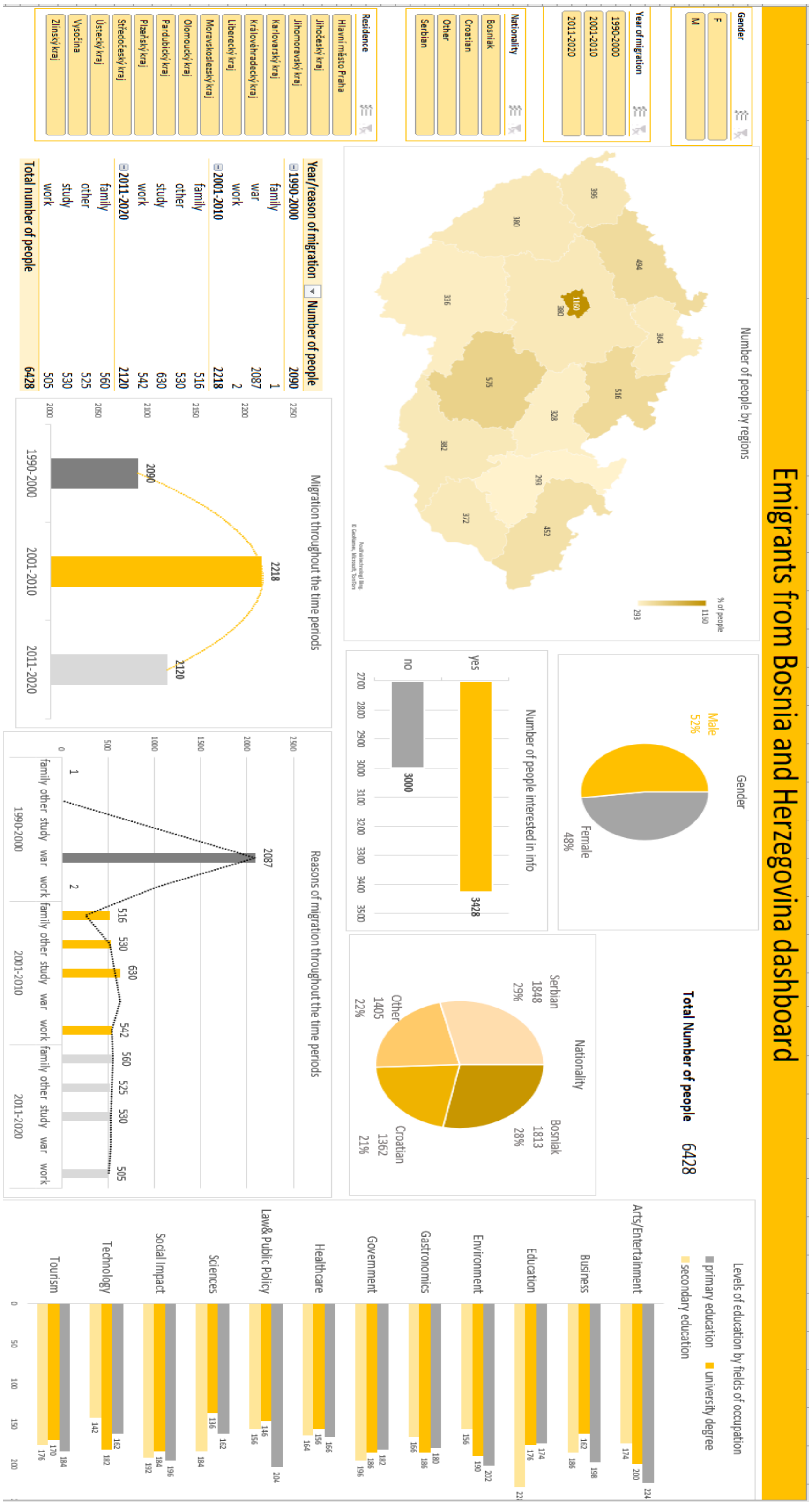


Annex 5 Source: Author

Annex F: Dashboard "Migrations throughout the years" in PBI

Annex 6 Source: Author

Annex G: Dashboard in Excel



Annex 7 Source: Author